

# Bd. 33 - Halbgruppen mit Nullelement

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# Kapitel 1

## Einführung

Dieser Band untersucht Halbgruppen mit einem absorbierenden Nullelement.

## Kapitel 2

# Halbgruppen mit Nullelement

### 2.1 Definition

Eine Halbgruppe kann ein absorbierendes Element besitzen. Wir nennen es *Nullelement*; der Ausdruck *Nullhalbgruppe* bleibt dagegen der spezielleren Situation vorbehalten, in der jedes Produkt gleich dem Nullelement ist.

**Definition 33.2.1.1 (Halbgruppe mit Nullelement).**

|                                    |                                            |
|------------------------------------|--------------------------------------------|
| <i>Mengen:</i>                     | $A, 0, x$                                  |
| <i>Axiome:</i>                     |                                            |
| <i>Halbgruppe:</i>                 | $\triangleright \text{HGr}(A, \star)$      |
| <i>Nullelement:</i>                | $\triangleright 0 \in A$                   |
| <i>Linksabsorption:</i>            | $x \in A \vdash 0 \star x = 0$             |
| <i>Rechtsabsorption:</i>           | $x \in A \vdash x \star 0 = 0$             |
| <i>Neues Symbol:</i>               |                                            |
| <i>Halbgruppe mit Nullelement:</i> | $\triangleright \text{HGr}_0(A, \star, 0)$ |

$$\text{HGr}_0(A, \star, 0)$$

**Axiom 33.2.1.1 (Zugriff auf die Halbgruppe).**

*Mengen:*  $A, 0$

$$\text{HGr}_0(A, \star, 0) \vdash \text{HGr}(A, \star)$$

**Axiom 33.2.1.2 (Zugriff auf das Nullelement).**

*Mengen:*  $A, 0$

$$\text{HGr}_0(A, \star, 0) \vdash 0 \in A$$

**Axiom 33.2.1.3 (Zugriff auf die Linksabsorption).**

**Mengen:**  $A, 0, x$

$$\text{HGr}_0(A, \star, 0), x \in A \vdash 0 \star x = 0$$

**Axiom 33.2.1.4 (Zugriff auf die Rechtsabsorption).**

**Mengen:**  $A, 0, x$

$$\text{HGr}_0(A, \star, 0), x \in A \vdash x \star 0 = 0$$

**Theorem 33.2.1.1 (Eindeutigkeit des Nullelements).**

**Mengen:**  $A, 0, z$

**Funktionen:**  $\star$

$$\text{HGr}_0(A, \star, 0), \text{HGr}_0(A, \star, z) \vdash 0 = z$$

|     |   |                             |                    |
|-----|---|-----------------------------|--------------------|
| 1   | 1 | $\text{HGr}_0(A, \star, 0)$ | $A$                |
| 2   | 2 | $\text{HGr}_0(A, \star, z)$ | $A$                |
| 1   | 3 | $0 \in A$                   | Ax. 33.2.1.2(1)    |
| 2   | 4 | $z \in A$                   | Ax. 33.2.1.2(2)    |
| 1,2 | 5 | $0 \star z = 0$             | Ax. 33.2.1.3(1, 4) |
| 1,2 | 6 | $0 \star z = z$             | Ax. 33.2.1.4(2, 3) |
| 1,2 | 7 | $0 = z$                     | Th. 2.9.1.4(5, 6)  |

## 2.2 Beispiele

**Theorem 33.2.2.1 (Die Multiplikation der natürlichen Zahlen besitzt ein Nullelement).**

**Mengen:**  $\mathbb{N}, 0, a$

$$\text{HGr}_0(\mathbb{N}, \cdot, 0)$$

|   |                                 |                 |
|---|---------------------------------|-----------------|
| 1 | $\text{HGr}(\mathbb{N}, \cdot)$ | Th. 28.2.5.4    |
| 2 | $0 \in \mathbb{N}$              | Ax. 10.3.1      |
| 3 | $a \in \mathbb{N}$              | $A$             |
| 3 | $0 \cdot a = 0$                 | Th. 10.4.7.4(3) |

$$\begin{array}{ll} 3 & 5 \ a \cdot 0 = 0 \\ & 6 \ \text{HGr}_0(\mathbb{N}, \cdot, 0) \end{array}$$

Th. 10.4.7.2(3)  
Def. 33.2.1.1(1, 2, 4, 5)

**Theorem 33.2.2.2 (Eine konstante Operation bildet eine Halbgruppe mit Nullelement).**

**Mengen:**  $A, p, x, y, z$

**Funktionen:**  $\star_0$

$$p \in A, \forall x, y \in A (x \star_0 y = p) \vdash \text{HGr}_0(A, \star_0, p)$$

*Beweis.*

Abgeschlossenheit der konstanten Operation:

$$\begin{array}{llll} 1 & 1 \ p \in A & A \\ 2 & 2 \ \forall x, y \in A (x \star_0 y = p) & A \\ 3 & 3 \ x \in A & A \\ 4 & 4 \ y \in A & A \\ 5 & 5 \ z \in A & A \\ 2,3,4 & 6 \ x \star_0 y = p & \rightarrow E(3, \rightarrow E(4, 2)) \\ 1,2,3,4 & 7 \ x \star_0 y \in A & = E(6, 1) \end{array}$$

Assoziativität und Halbgruppenstruktur:

$$\begin{array}{llll} 1,2,3,4,5 & 8 \ (x \star_0 y) \star_0 z = p & \rightarrow E(7, \rightarrow E(5, 2)) \\ 2,4,5 & 9 \ y \star_0 z = p & \rightarrow E(4, \rightarrow E(5, 2)) \\ 1,2,4,5 & 10 \ y \star_0 z \in A & = E(9, 1) \\ 1,2,3,4,5 & 11 \ x \star_0 (y \star_0 z) = p & \rightarrow E(3, \rightarrow E(10, 2)) \\ 1,2,3,4,5 & 12 \ p = x \star_0 (y \star_0 z) & \text{Th. 2.9.1.1(11)} \\ 1,2,3,4,5 & 13 \ (x \star_0 y) \star_0 z = x \star_0 (y \star_0 z) & \text{Tr. (8, 12)} \\ 1,2 & 14 \ \text{HGr}(A, \star_0) & \text{Def. 28.2.1.1(7, 13)} \end{array}$$

Beidseitige Nullabsorption:

$$\begin{array}{llll} 1,2,3 & 15 \ p \star_0 x = p & \rightarrow E(1, \rightarrow E(3, 2)) \\ 1,2,3 & 16 \ x \star_0 p = p & \rightarrow E(3, \rightarrow E(1, 2)) \\ 1,2 & 17 \ \text{HGr}_0(A, \star_0, p) & \text{Def. 33.2.1.1(14, 1, 15, 16)} \end{array}$$

□

## 2.3 Annihilatoren

Annihilatoren werden für Halbgruppen mit Nullelement und relativ zu einer Teilmenge definiert. Links und rechts müssen getrennt werden, solange keine Kommutativität vorausgesetzt ist.

**Definition 33.2.3.1 (Linksannihilator einer Teilmenge).**

*Mengen:*  $A, X, 0, a, x$

*Funktionen:*  $\star$

*Neues Symbol:*

*Linksannihilator:*  $\triangleright \text{Ann}_{A,\star,0}^{\ell}(X)$

$$\begin{aligned} & \text{HGr}_0(A, \star, 0), X \subseteq A \vdash \\ & \text{Ann}_{A,\star,0}^{\ell}(X) := \{a \in A \mid \forall x \in X (a \star x = 0)\} \end{aligned}$$

**Definition 33.2.3.2 (Rechtsannihilator einer Teilmenge).**

*Mengen:*  $A, X, 0, a, x$

*Funktionen:*  $\star$

*Neues Symbol:*

*Rechtsannihilator:*  $\triangleright \text{Ann}_{A,\star,0}^r(X)$

$$\begin{aligned} & \text{HGr}_0(A, \star, 0), X \subseteq A \vdash \\ & \text{Ann}_{A,\star,0}^r(X) := \{a \in A \mid \forall x \in X (x \star a = 0)\} \end{aligned}$$

**Definition 33.2.3.3 (Zweiseitiger Annihilator einer Teilmenge).**

*Mengen:*  $A, X, 0$   
*Funktionen:*  $\star$   
*Neues Symbol:*  
*Zweiseitiger Annihilator:*  $\triangleright \text{Ann}_{A,\star,0}(X)$

$$\text{HGr}_0(A, \star, 0), X \subseteq A \vdash$$

$$\text{Ann}_{A,\star,0}(X) := \text{Ann}_{A,\star,0}^\ell(X) \cap \text{Ann}_{A,\star,0}^r(X)$$

Damit gilt ausgeschrieben

$$\text{Ann}_{A,\star,0}(X) = \{a \in A \mid \forall x \in X (a \star x = 0 \wedge x \star a = 0)\}.$$

Für  $X = A$  heißt  $\text{Ann}_{A,\star,0}(A)$  der *Totalannihilator*. Bei Kommutativität stimmen Links-, Rechts- und zweiseitiger Annihilator überein. Insbesondere gehört das Nullelement zu jedem dieser Annihilatoren.

## 2.4 Morphismen

**Definition 33.2.4.1 (Homomorphismus von Halbgruppen mit Nullelement).**

*Mengen:*  $A, B, 0_A, 0_B$   
*Funktionen:*  $f, \star, \diamond$   
*Axiome:*  
*Quellstruktur:*  $\triangleright \text{HGr}_0(A, \star, 0_A)$   
*Zielstruktur:*  $\triangleright \text{HGr}_0(B, \diamond, 0_B)$   
*Halbgruppenhomomorphismus:*  $\triangleright \text{HGrHom}(f; A, \star; B, \diamond)$   
*Nullerhaltung:*  $f(0_A) = 0_B$   
*Neues Symbol:*  
*Homomorphismus:*  $\triangleright \text{HGr}_0\text{Hom}(f; A, \star, 0_A; B, \diamond, 0_B)$

$$\text{HGr}_0\text{Hom}(f; A, \star, 0_A; B, \diamond, 0_B)$$

**Axiom 33.2.4.1 (Quellstruktur).**

*Mengen:*  $A, B, 0_A, 0_B$   
*Funktionen:*  $f, \star, \diamond$

$$\text{HGr}_0\text{Hom}(f; A, \star, 0_A; B, \diamond, 0_B) \vdash \text{HGr}_0(A, \star, 0_A)$$

**Axiom 33.2.4.2 (Zielstruktur).**

*Mengen:*  $A, B, 0_A, 0_B$   
*Funktionen:*  $f, \star, \diamond$

$$\text{HGr}_0\text{Hom}(f; A, \star, 0_A; B, \diamond, 0_B) \vdash \text{HGr}_0(B, \diamond, 0_B)$$

**Axiom 33.2.4.3 (Zugriff auf den Halbgruppenhomomorphismus).**

*Mengen:*  $A, B, 0_A, 0_B$

*Funktionen:*  $f, \star, \diamond$

$$\text{HGr}_0\text{Hom}(f; A, \star, 0_A; B, \diamond, 0_B) \vdash \text{HGrHom}(f; A, \star; B, \diamond)$$

**Axiom 33.2.4.4 (Zugriff auf die Nullerhaltung).**

*Mengen:*  $A, B, 0_A, 0_B$

*Funktionen:*  $f, \star, \diamond$

$$\text{HGr}_0\text{Hom}(f; A, \star, 0_A; B, \diamond, 0_B) \vdash f(0_A) = 0_B$$

**Definition 33.2.4.2 (Isomorphismus von Halbgruppen mit Nullelement).**

*Mengen:*  $A, B, 0_A, 0_B$

*Funktionen:*  $f, \star, \diamond$

*Axiome:*

*Homomorphismus:*  $\triangleright \text{HGr}_0\text{Hom}(f; A, \star, 0_A; B, \diamond, 0_B)$

*Bijektivität:*  $\triangleright f: A \xrightarrow{\sim} B$

*Neues Symbol:*

*Isomorphismus:*  $\triangleright \text{HGr}_0\text{Iso}(f; A, \star, 0_A; B, \diamond, 0_B)$

$$\text{HGr}_0\text{Iso}(f; A, \star, 0_A; B, \diamond, 0_B)$$

**Axiom 33.2.4.5 (Homomorphismuszugriff).**

*Mengen:*  $A, B, 0_A, 0_B$

*Funktionen:*  $f, \star, \diamond$

$$\text{HGr}_0\text{Iso}(f; A, \star, 0_A; B, \diamond, 0_B) \vdash \text{HGr}_0\text{Hom}(f; A, \star, 0_A; B, \diamond, 0_B)$$

**Axiom 33.2.4.6 (Bijektivitätszugriff).**

*Mengen:*  $A, B, 0_A, 0_B$

*Funktionen:*  $f, \star, \diamond$

$$\text{HGr}_0\text{Iso}(f; A, \star, 0_A; B, \diamond, 0_B) \vdash f: A \xrightarrow{\sim} B$$

## Kapitel 3

# Nullproduktprofile

Die Nullprodukte einer Halbgruppe lassen sich bereits in ihrer Potenzhalbgruppe erkennen. Dabei werden keine Einermengen erkannt: zunächst erhält man Klassen mit gleichem Verhalten gegenüber allen linken und rechten Nulltests. Bei endlichen Trägern bestimmen die Größen unterer Mengen, wie viele Grundelemente in jeder dieser Klassen liegen.

Wir schreiben  $K_S = \mathcal{P}(S) \setminus \{\emptyset\}$ ,  $z_S = \{0_S\}$  und entsprechend  $K_T = \mathcal{P}(T) \setminus \{\emptyset\}$ ,  $z_T = \{0_T\}$ . Nebeneinanderschreiben bezeichnet bei Grundelementen die jeweilige Halbgruppenoperation und bei Teilmengen das Mengenprodukt aus Def. 28.2.4.1. Die Abkürzungen

$$\begin{aligned} H_S &= \text{HGr}_0(S, \star, 0_S), & H_T &= \text{HGr}_0(T, \diamond, 0_T), \\ I &= \text{HGrIso}(\Phi; K_S, \star_{\mathcal{P}}; K_T, \diamond_{\mathcal{P}}). \end{aligned}$$

werden in Aussagen und Tabellen nur zur Kürzung wiederholter Voraussetzungen verwendet.  $I$  enthält insbesondere die Bijektivität von  $\Phi$ .

### 3.1 Einermengen und Nullprodukte

**Theorem 33.3.1.1 (Einermengen sind nichtleere Teilmengen).**

**Mengen:**  $S, a$

$$a \in S \vdash \{a\} \in \mathcal{P}(S) \setminus \{\emptyset\}$$

|   |   |                                                    |                                  |
|---|---|----------------------------------------------------|----------------------------------|
| 1 | 1 | $a \in S$                                          | $A$                              |
| 1 | 2 | $\{a\} \subseteq S$                                | Th. 3.11.3.15(1)                 |
|   | 3 | $a \in \{a\}$                                      | Th. 3.11.3.8                     |
|   | 4 | $\{a\} \neq \emptyset$                             | Th. 3.6.3.5(3)                   |
| 1 | 5 | $\{a\} \in \mathcal{P}(S) \setminus \{\emptyset\}$ | Th. 3.16.3.1( $\wedge I(2, 4)$ ) |

**Theorem 33.3.1.2 (Konstante Produkte nichtleerer Teilmengen).**

**Mengen:**  $S, A, B, a, b, d, w$

$$\text{HGr}(S, \star), A, B \in K_S \vdash (AB = \{d\} \leftrightarrow \forall a \in A \forall b \in B (ab = d))$$

*Beweis.*

Hinrichtung:

|           |    |                                                                   |                                                               |
|-----------|----|-------------------------------------------------------------------|---------------------------------------------------------------|
| 1         | 1  | $\text{HGr}(S, \star)$                                            | $A$                                                           |
| 2         | 2  | $A, B \in K_S$                                                    | $A$                                                           |
| 3         | 3  | $AB = \{d\}$                                                      | $A$                                                           |
| 4         | 4  | $a \in A$                                                         | $A$                                                           |
| 5         | 5  | $b \in B$                                                         | $A$                                                           |
| 1,2,4,5   | 6  | $ab \in AB$                                                       | Th. 28.2.4.2( $H2$ )(1, 2, 4, 5)                              |
| 1,2,3,4,5 | 7  | $ab \in \{d\}$                                                    | $= E(3, 6)$                                                   |
| 1,2,3,4,5 | 8  | $ab = d$                                                          | Th. 3.11.3.9(7)                                               |
| 1,2,3     | 9  | $\forall a \in A \forall b \in B (ab = d)$                        | $\forall I(\rightarrow I(4, \forall I(\rightarrow I(5, 8))))$ |
| 1,2       | 10 | $AB = \{d\} \rightarrow \forall a \in A \forall b \in B (ab = d)$ | $\rightarrow I(3, 9)$                                         |

Rückrichtung:

|           |    |                                                                       |                                                                  |
|-----------|----|-----------------------------------------------------------------------|------------------------------------------------------------------|
| 11        | 11 | $\forall a \in A \forall b \in B (ab = d)$                            | $A$                                                              |
| 12        | 12 | $w \in AB$                                                            | $A$                                                              |
| 1,2,12    | 13 | $\exists a \in A \exists b \in B (w = ab)$                            | Th. 28.2.4.2(H6)(1, 2, 12)                                       |
| 14        | 14 | $a \in A \wedge b \in B \wedge w = ab$                                | $A$                                                              |
| 11,14     | 15 | $ab = d$                                                              | $\rightarrow E(\forall E(\rightarrow E(\forall E(11), 14)), 14)$ |
| 11,14     | 16 | $w = d$                                                               | Tr. $(\wedge E2(\wedge E2(14)), 15)$                             |
| 11,14     | 17 | $w \in \{d\}$                                                         | Th. 3.11.3.9(16)                                                 |
| 1,2,11,12 | 18 | $w \in \{d\}$                                                         | $\exists E^*(13, 14-17)$                                         |
| 1,2,11    | 19 | $AB \subseteq \{d\}$                                                  | Th. 3.5.1.1([12]:18)                                             |
| 1,2       | 20 | $AB \neq \emptyset$                                                   | Th. 28.2.4.2(H4)(1, 2)                                           |
| 1,2       | 21 | $\exists w w \in AB$                                                  | Th. 3.6.3.7(20)                                                  |
| 22        | 22 | $w \in AB$                                                            | $A$                                                              |
| 1,2,11,22 | 23 | $w \in \{d\}$                                                         | Th. 3.5.2.6(19, 22)                                              |
| 1,2,11,22 | 24 | $w = d$                                                               | Th. 3.11.3.9(23)                                                 |
| 1,2,11,22 | 25 | $d \in AB$                                                            | $= E(24, 22)$                                                    |
| 1,2,11    | 26 | $d \in AB$                                                            | $\exists E(21, [22]:25)$                                         |
| 1,2,11    | 27 | $\{d\} \subseteq AB$                                                  | Th. 3.11.3.15(26)                                                |
| 1,2,11    | 28 | $AB = \{d\}$                                                          | Th. 3.5.3.5(19, 27)                                              |
| 1,2       | 29 | $(\forall a \in A \forall b \in B (ab = d)) \rightarrow AB = \{d\}$   | $\rightarrow I(11, 28)$                                          |
| 1,2       | 30 | $AB = \{d\} \leftrightarrow \forall a \in A \forall b \in B (ab = d)$ | $\leftrightarrow I(10, 29)$                                      |

□

**Theorem 33.3.1.3 (Produkt zweier Einermengen).**

**Mengen:**  $S, a, b, x, y$

$$\text{HGr}(S, \star), a, b \in S \vdash \{a\}\{b\} = \{ab\}$$

|   |   |                        |                 |
|---|---|------------------------|-----------------|
| 1 | 1 | $\text{HGr}(S, \star)$ | $A$             |
| 2 | 2 | $a, b \in S$           | $A$             |
| 2 | 3 | $\{a\}, \{b\} \in K_S$ | Th. 33.3.1.1(2) |
| 4 | 4 | $x \in \{a\}$          | $A$             |
| 5 | 5 | $y \in \{b\}$          | $A$             |
| 4 | 6 | $x = a$                | Th. 3.11.3.9(4) |

|     |    |                                                     |                                                                     |
|-----|----|-----------------------------------------------------|---------------------------------------------------------------------|
| 5   | 7  | $y = b$                                             | Th. 3.11.3.9(5)                                                     |
| 4,5 | 8  | $xy = ab$                                           | $= E(7, = E(6, = I))$                                               |
|     | 9  | $\forall x \in \{a\} \forall y \in \{b\} (xy = ab)$ | $\forall I(\rightarrow I(4, \forall I(\rightarrow I$<br>$(5, 8))))$ |
| 1,2 | 10 | $\{a\}\{b\} = \{ab\}$                               | Th. 33.3.1.2(1, 3, 9)                                               |

**Theorem 33.3.1.4 (Die nichtleere Potenzhalbgruppe hat eine Null).**

**Mengen:**  $S, 0_S, A, x$

$$H_S \vdash \text{HGr}_0(K_S, \star_P, z_S)$$

|          |    |                                                |                                                                                          |
|----------|----|------------------------------------------------|------------------------------------------------------------------------------------------|
| 1        | 1  | $H_S$                                          | $A$                                                                                      |
| 1        | 2  | $\text{HGr}(S, \star)$                         | Ax. 33.2.1.1(1)                                                                          |
| 1        | 3  | $0_S \in S$                                    | Ax. 33.2.1.2(1)                                                                          |
| 1        | 4  | $\text{HGr}(K_S, \star_P)$                     | Th. 28.2.4.2(2)                                                                          |
| 1        | 5  | $z_S \in K_S$                                  | Th. 33.3.1.1(3)                                                                          |
| 6        | 6  | $A \in K_S$                                    | $A$                                                                                      |
| 7        | 7  | $x \in A$                                      | $A$                                                                                      |
| 6,7      | 8  | $x \in S$                                      | Th. 28.2.4.2(H1)(6, 7)                                                                   |
| 1,6,7    | 9  | $0_S x = 0_S \wedge x 0_S = 0_S$               | $\wedge I(\text{Ax. 33.2.1.3}(1, 8), \text{Ax. 33.2.1.4}(1, 8))$                         |
| 10       | 10 | $u \in z_S$                                    | $A$                                                                                      |
| 10       | 11 | $u = 0_S$                                      | Th. 3.11.3.9(10)                                                                         |
| 1,6,7,10 | 12 | $ux = 0_S \wedge xu = 0_S$                     | $= E(\text{Th. 2.9.1.1}(11), 9)$                                                         |
| 1,6      | 13 | $\forall u \in z_S \forall x \in A (ux = 0_S)$ | $\forall I(\rightarrow I(10, \forall I(\rightarrow I(7, \wedge E1(12))))$                |
| 1,6      | 14 | $\forall x \in A \forall u \in z_S (xu = 0_S)$ | $\forall I(\rightarrow I(7, \forall I(\rightarrow I(10, \wedge E2(12))))$                |
| 1,6      | 15 | $z_S A = z_S$                                  | Th. 33.3.1.2(2, 5, 6, 13)                                                                |
| 1,6      | 16 | $A z_S = z_S$                                  | Th. 33.3.1.2(2, 6, 5, 14)                                                                |
| 1        | 17 | $\text{HGr}_0(K_S, \star_P, z_S)$              | Def. 33.2.1.1(4, 5, $\forall I(\rightarrow I(6, 15)), \forall I(\rightarrow I(6, 16))$ ) |

**Theorem 33.3.1.5 (Isomorphismen erhalten die Null der Potenzhalbgruppe).**

**Mengen:**  $S, T, 0_S, 0_T, Y$

$$H_S, H_T, I \vdash \Phi(z_S) = z_T$$

|       |    |                                                                              |                                                 |
|-------|----|------------------------------------------------------------------------------|-------------------------------------------------|
| 1     | 1  | $H_S$                                                                        | $A$                                             |
| 2     | 2  | $H_T$                                                                        | $A$                                             |
| 3     | 3  | $I$                                                                          | $A$                                             |
| 1     | 4  | $\text{HGr}_0(K_S, \star_P, z_S)$                                            | Th. 33.3.1.4(1)                                 |
| 2     | 5  | $\text{HGr}_0(K_T, \diamond_P, z_T)$                                         | Th. 33.3.1.4(2)                                 |
| 1     | 6  | $z_S \in K_S$                                                                | Ax. 33.2.1.2(4)                                 |
| 3     | 7  | $\Phi : K_S \rightarrow K_T$                                                 | Th. 28.2.15.2(H3)<br>(3)                        |
| 1,3   | 8  | $\Phi(z_S) \in K_T$                                                          | Th. 5.2.3.9(7, 6)                               |
| 1     | 9  | $\forall X \in K_S (z_S X = z_S \wedge X z_S = z_S)$                         | Def. 33.2.1.1(4)                                |
| 1,3   | 10 | $\forall Y \in K_T (\Phi(z_S) Y = \Phi(z_S) \wedge Y \Phi(z_S) = \Phi(z_S))$ | Th. 28.2.15.22(H1)<br>(3, $\wedge I(6, 6), 9$ ) |
| 3     | 11 | $\text{HGr}(K_T, \diamond_P)$                                                | Th. 28.2.15.2(H2)<br>(3)                        |
| 1,3   | 12 | $\text{HGr}_0(K_T, \diamond_P, \Phi(z_S))$                                   | Def. 33.2.1.1(11, 8,<br>10)                     |
| 1,2,3 | 13 | $\Phi(z_S) = z_T$                                                            | Th. 33.2.1.1(12, 5)                             |

**Theorem 33.3.1.6 (Linke Zerlegung eines konstanten Mengenprodukts).**

**Mengen:**  $S, A, X, a, x, u, d$

$$\text{HGr}(S, \star), A, X \in K_S \vdash (AX = \{d\} \leftrightarrow \forall a \in A \{a\}X = \{d\})$$

*Beweis.*

Hinrichtung:

|       |   |                                                                       |                    |
|-------|---|-----------------------------------------------------------------------|--------------------|
| 1     | 1 | $\text{HGr}(S, \star)$                                                | $A$                |
| 2     | 2 | $A, X \in K_S$                                                        | $A$                |
| 3     | 3 | $AX = \{d\}$                                                          | $A$                |
| 1,2   | 4 | $AX = \{d\} \leftrightarrow \forall a \in A \forall x \in X (ax = d)$ | Th. 33.3.1.2(1, 2) |
| 1,2,3 | 5 | $\forall a \in A \forall x \in X (ax = d)$                            | Th. 2.2.4.1(4, 3)  |
| 6     | 6 | $a \in A$                                                             | $A$                |

|                 |    |                                                         |                                                                    |
|-----------------|----|---------------------------------------------------------|--------------------------------------------------------------------|
| 7               | 7  | $u \in \{a\}$                                           | $A$                                                                |
| 8               | 8  | $x \in X$                                               | $A$                                                                |
| 7               | 9  | $u = a$                                                 | Th. 3.11.3.9(7)                                                    |
| 1,2,3,6,8       | 10 | $ax = d$                                                | $\rightarrow E(\forall E(\rightarrow E(\forall E$<br>(5), 6)), 8)  |
| 1,2,3,6,7,<br>8 | 11 | $ux = d$                                                | $= E(\text{Th. 2.9.1.1(9),}$<br>10)                                |
| 1,2,3,6         | 12 | $\forall u \in \{a\} \forall x \in X (ux = d)$          | $\forall I(\rightarrow I(7, \forall I(\rightarrow I$<br>(8, 11)))) |
| 2,6             | 13 | $a \in S$                                               | Th. 28.2.4.2( $H1$ )(2,<br>6)                                      |
| 2,6             | 14 | $\{a\} \in K_S$                                         | Th. 33.3.1.1(13)                                                   |
| 1,2,3,6         | 15 | $\{a\}X = \{d\}$                                        | Th. 33.3.1.2(1, 2, 14,<br>12)                                      |
| 1,2,3           | 16 | $\forall a \in A \{a\}X = \{d\}$                        | $\forall I(\rightarrow I(6, 15))$                                  |
| 1,2             | 17 | $AX = \{d\} \rightarrow \forall a \in A \{a\}X = \{d\}$ | $\rightarrow I(3, 16)$                                             |

Rückrichtung:

|                  |    |                                                             |                                                                      |
|------------------|----|-------------------------------------------------------------|----------------------------------------------------------------------|
| 18               | 18 | $a \in A$                                                   | $A$                                                                  |
| 19               | 19 | $x \in X$                                                   | $A$                                                                  |
| 20               | 20 | $\forall a \in A \{a\}X = \{d\}$                            | $A$                                                                  |
| 18,20            | 21 | $\{a\}X = \{d\}$                                            | $\rightarrow E(\forall E(20), 18)$                                   |
| 2,18             | 22 | $\{a\} \in K_S$                                             | Th. 33.3.1.1(<br>Th. 28.2.4.2( $H1$ )(2,<br>18))                     |
|                  | 23 | $a \in \{a\}$                                               | Th. 3.11.3.8                                                         |
| 1,2,18,<br>19,20 | 24 | $ax = d$                                                    | Th. 33.3.1.2(1, 2, 22,<br>21, 23, 19)                                |
| 1,2,20           | 25 | $\forall a \in A \forall x \in X (ax = d)$                  | $\forall I(\rightarrow I(18, \forall I(\rightarrow I$<br>(19, 24)))) |
| 1,2,20           | 26 | $AX = \{d\}$                                                | Th. 33.3.1.2(1, 2, 25)                                               |
| 1,2              | 27 | $(\forall a \in A \{a\}X = \{d\}) \rightarrow AX = \{d\}$   | $\rightarrow I(20, 26)$                                              |
| 1,2              | 28 | $AX = \{d\} \leftrightarrow \forall a \in A \{a\}X = \{d\}$ | $\leftrightarrow I(17, 27)$                                          |

□

**Theorem 33.3.1.7 (Rechte Zerlegung eines konstanten Mengenprodukts).**

**Mengen:**  $S, A, X, a, x, u, d$

$$\text{HGr}(S, \star), A, X \in K_S \vdash (XA = \{d\} \leftrightarrow \forall a \in A X\{a\} = \{d\})$$

*Beweis.*

Hinrichtung:

|             |    |                                                                       |                                                                |
|-------------|----|-----------------------------------------------------------------------|----------------------------------------------------------------|
| 1           | 1  | $\text{HGr}(S, \star)$                                                | $A$                                                            |
| 2           | 2  | $A, X \in K_S$                                                        | $A$                                                            |
| 3           | 3  | $XA = \{d\}$                                                          | $A$                                                            |
| 1,2         | 4  | $XA = \{d\} \leftrightarrow \forall x \in X \forall a \in A (xa = d)$ | Th. 33.3.1.2(1, 2)                                             |
| 1,2,3       | 5  | $\forall x \in X \forall a \in A (xa = d)$                            | Th. 2.2.4.1(4, 3)                                              |
| 6           | 6  | $a \in A$                                                             | $A$                                                            |
| 7           | 7  | $u \in \{a\}$                                                         | $A$                                                            |
| 8           | 8  | $x \in X$                                                             | $A$                                                            |
| 7           | 9  | $u = a$                                                               | Th. 3.11.3.9(7)                                                |
| 1,2,3,6,8   | 10 | $xa = d$                                                              | $\rightarrow E(\forall E(\rightarrow E(\forall E(5), 8)), 6)$  |
| 1,2,3,6,7,8 | 11 | $xu = d$                                                              | $= E(\text{Th. 2.9.1.1}(9), 10)$                               |
| 1,2,3,6     | 12 | $\forall x \in X \forall u \in \{a\} (xu = d)$                        | $\forall I(\rightarrow I(8, \forall I(\rightarrow I(7, 11))))$ |
| 2,6         | 13 | $a \in S$                                                             | Th. 28.2.4.2(H1)(2, 6)                                         |
| 2,6         | 14 | $\{a\} \in K_S$                                                       | Th. 33.3.1.1(13)                                               |
| 1,2,3,6     | 15 | $X\{a\} = \{d\}$                                                      | Th. 33.3.1.2(1, 2, 14, 12)                                     |
| 1,2,3       | 16 | $\forall a \in A X\{a\} = \{d\}$                                      | $\forall I(\rightarrow I(6, 15))$                              |
| 1,2         | 17 | $XA = \{d\} \rightarrow \forall a \in A X\{a\} = \{d\}$               | $\rightarrow I(3, 16)$                                         |

Rückrichtung:

|       |    |                                  |                                       |
|-------|----|----------------------------------|---------------------------------------|
| 18    | 18 | $a \in A$                        | $A$                                   |
| 19    | 19 | $x \in X$                        | $A$                                   |
| 20    | 20 | $\forall a \in A X\{a\} = \{d\}$ | $A$                                   |
| 18,20 | 21 | $X\{a\} = \{d\}$                 | $\rightarrow E(\forall E(20), 18)$    |
| 2,18  | 22 | $\{a\} \in K_S$                  | Th. 33.3.1.1(Th. 28.2.4.2(H1)(2, 18)) |

|                  |    |                                                             |                                                                      |
|------------------|----|-------------------------------------------------------------|----------------------------------------------------------------------|
|                  | 23 | $a \in \{a\}$                                               | Th. 3.11.3.8                                                         |
| 1,2,18,<br>19,20 | 24 | $xa = d$                                                    | Th. 33.3.1.2(1, 2, 22,<br>21, 23, 19)                                |
| 1,2,20           | 25 | $\forall x \in X \forall a \in A (xa = d)$                  | $\forall I(\rightarrow I(19, \forall I(\rightarrow I$<br>(18, 24)))) |
| 1,2,20           | 26 | $XA = \{d\}$                                                | Th. 33.3.1.2(1, 2, 25)                                               |
| 1,2              | 27 | $(\forall a \in A X\{a\} = \{d\}) \rightarrow XA = \{d\}$   | $\rightarrow I(20, 26)$                                              |
| 1,2              | 28 | $XA = \{d\} \leftrightarrow \forall a \in A X\{a\} = \{d\}$ | $\leftrightarrow I(17, 27)$                                          |

□

## 3.2 Präordnung und Quotientenordnung

**Definition 33.3.2.1 (Vergleich der Nullproduktprofile).**

**Halbgruppe mit Null:**  $\triangleright H_S$   
**Mengen:**  $A, B, X, S, 0_S$

$$A \preceq_S B := \forall X \in K_S ((BX = z_S \rightarrow AX = z_S) \wedge (XB = z_S \rightarrow XA = z_S))$$

**Definition 33.3.2.2 (Gleiches Nullproduktprofil).**

**Halbgruppe mit Null:**  $\triangleright H_S$   
**Mengen:**  $A, B, S, 0_S$

$$A \sim_S B := (A \preceq_S B \wedge B \preceq_S A)$$

**Theorem 33.3.2.1 (Nullproduktprofile bilden eine Präordnung).**

**Mengen:**  $S, A, B, C, X$

$$H_S \vdash \text{PrO}(K_S, \preceq_S)$$

*Beweis.*

**Reflexivität**

Th. 33.3.2.1(H1)

$$H_S, A \in K_S \vdash A \preceq_S A$$

|   |   |             |     |
|---|---|-------------|-----|
| 1 | 1 | $H_S$       | $A$ |
| 2 | 2 | $A \in K_S$ | $A$ |
| 3 | 3 | $X \in K_S$ | $A$ |

|     |   |                                                                                              |                                               |
|-----|---|----------------------------------------------------------------------------------------------|-----------------------------------------------|
| 4   | 4 | $AX = z_S$                                                                                   | $A$                                           |
|     | 5 | $AX = z_S \rightarrow AX = z_S$                                                              | $\rightarrow I(4, 4)$                         |
| 6   | 6 | $XA = z_S$                                                                                   | $A$                                           |
|     | 7 | $XA = z_S \rightarrow XA = z_S$                                                              | $\rightarrow I(6, 6)$                         |
|     | 8 | $\forall X \in K_S ((AX = z_S \rightarrow AX = z_S) \wedge (XA = z_S \rightarrow XA = z_S))$ | $\forall I(\rightarrow I(3, \wedge I(5, 7)))$ |
| 1,2 | 9 | $A \preceq_S A$                                                                              | Def. 33.3.2.1(1, 2, 8)                        |

**Transitivität**

Th. 33.3.2.1(H2)

$$H_S, A, B, C \in K_S, A \preceq_S B, B \preceq_S C \vdash A \preceq_S C$$

|            |    |                                                                                              |                                                                |
|------------|----|----------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| 1          | 1  | $H_S$                                                                                        | $A$                                                            |
| 2          | 2  | $A, B, C \in K_S$                                                                            | $A$                                                            |
| 3          | 3  | $A \preceq_S B$                                                                              | $A$                                                            |
| 4          | 4  | $B \preceq_S C$                                                                              | $A$                                                            |
| 5          | 5  | $X \in K_S$                                                                                  | $A$                                                            |
| 1,3,5      | 6  | $(BX = z_S \rightarrow AX = z_S) \wedge (XB = z_S \rightarrow XA = z_S)$                     | Def. 33.3.2.1(1, 3, 5)                                         |
| 1,4,5      | 7  | $(CX = z_S \rightarrow BX = z_S) \wedge (XC = z_S \rightarrow XB = z_S)$                     | Def. 33.3.2.1(1, 4, 5)                                         |
| 8          | 8  | $CX = z_S$                                                                                   | $A$                                                            |
| 1,3,4,5,8  | 9  | $AX = z_S$                                                                                   | $\rightarrow E(\wedge E1(6), \rightarrow E(\wedge E1(7), 8))$  |
| 1,3,4,5    | 10 | $CX = z_S \rightarrow AX = z_S$                                                              | $\rightarrow I(8, 9)$                                          |
| 11         | 11 | $XC = z_S$                                                                                   | $A$                                                            |
| 1,3,4,5,11 | 12 | $XA = z_S$                                                                                   | $\rightarrow E(\wedge E2(6), \rightarrow E(\wedge E2(7), 11))$ |
| 1,3,4,5    | 13 | $XC = z_S \rightarrow XA = z_S$                                                              | $\rightarrow I(11, 12)$                                        |
| 1,3,4      | 14 | $\forall X \in K_S ((CX = z_S \rightarrow AX = z_S) \wedge (XC = z_S \rightarrow XA = z_S))$ | $\forall I(\rightarrow I(5, \wedge I(10, 13)))$                |
| 1,2,3,4    | 15 | $A \preceq_S C$                                                                              | Def. 33.3.2.1(1, 2, 14)                                        |

Präordnung:

|     |   |                                      |                        |
|-----|---|--------------------------------------|------------------------|
| 1   | 1 | $H_S$                                | $A$                    |
| 2   | 2 | $A \in K_S$                          | $A$                    |
| 1,2 | 3 | $A \preceq_S A$                      | Th. 33.3.2.1(H1)(1, 2) |
| 4   | 4 | $B \in K_S$                          | $A$                    |
| 5   | 5 | $C \in K_S$                          | $A$                    |
| 6   | 6 | $A \preceq_S B \wedge B \preceq_S C$ | $A$                    |

1,2,4,5,6    7  $A \preceq_S C$

Th. 33.3.2.1(H2)(1,  
2, 4, 5, 6)

1    8  $\text{PrO}(K_S, \preceq_S)$

Def. 12.4.1( $\forall I(\rightarrow I$   
(2, 3)),  $\forall I(\rightarrow I(2, \forall I$   
( $\rightarrow I(4, \forall I(\rightarrow I(5,$   
 $\rightarrow I(6, 7))))))$ )

□

**Theorem 33.3.2.2 (Die Profilrelation ist eine Äquivalenzrelation).**

**Mengen:**  $S, 0_S$

$$H_S \vdash \text{EqRel}(K_S, \sim_S)$$

1    1  $H_S$

$A$

1    2  $\text{PrO}(K_S, \preceq_S)$

Th. 33.3.2.1(1)

1    3  $\text{EqRel}(K_S, \sim_S)$

Th. 12.4.1(2)

**Definition 33.3.2.3 (Profilquotient und Singleton-Profilfasern).**

**Halbgruppe mit Null:**  $\triangleright H_S$

**Mengen:**  $S, A, p, Q_S, L_p, D_p, C_p$

**Quotientenordnung:**  $\leq_S$  aus Def. 12.4.3

$$\begin{aligned} Q_S &:= K_S / \sim_S, & [A]_S &:= [A]_{\sim_S, K_S}, \\ L_p &:= \{A \in K_S \mid [A]_S \leq_S p\}, & D_p &:= \{s \in S \mid [\{s\}]_S \leq_S p\}, \\ C_p &:= \{s \in S \mid [\{s\}]_S = p\}. \end{aligned}$$

**Theorem 33.3.2.3 (Die Menge der Profile ist partiell geordnet).**

**Mengen:**  $S, Q_S$

$$H_S \vdash \text{PartOrd}(Q_S, \leq_S)$$

1    1  $H_S$

$A$

1    2  $\text{PrO}(K_S, \preceq_S)$

Th. 33.3.2.1(1)

1    3  $\text{PartOrd}(Q_S, \leq_S)$

Th. 12.4.4(2)

**Theorem 33.3.2.4 (Vergleich der Profile von Teilmengen).**

**Mengen:**  $S, A, B$

$$H_S, A, B \in K_S \vdash ([A]_S \leq_S [B]_S \leftrightarrow A \preceq_S B)$$

|     |   |                                                    |                  |
|-----|---|----------------------------------------------------|------------------|
| 1   | 1 | $H_S$                                              | $A$              |
| 2   | 2 | $A, B \in K_S$                                     | $A$              |
| 1   | 3 | $\text{PrO}(K_S, \preceq_S)$                       | Th. 33.3.2.1(1)  |
| 1,2 | 4 | $[A]_S \leq_S [B]_S \leftrightarrow A \preceq_S B$ | Th. 12.4.3(3, 2) |

**Theorem 33.3.2.5 (Ein Profilvergleich zerfällt in Singletonvergleiche).**

**Mengen:**  $S, A, B, X, a$

$$H_S, A, B \in K_S \vdash (A \preceq_S B \leftrightarrow \forall a \in A \{a\} \preceq_S B)$$

*Beweis.*

Von der Menge zu ihren Elementen:

|              |    |                                                                          |                                                                            |
|--------------|----|--------------------------------------------------------------------------|----------------------------------------------------------------------------|
| 1            | 1  | $H_S$                                                                    | $A$                                                                        |
| 2            | 2  | $A, B \in K_S$                                                           | $A$                                                                        |
| 1            | 3  | $\text{HGr}(S, \star)$                                                   | Ax. 33.2.1.1(1)                                                            |
| 4            | 4  | $A \preceq_S B$                                                          | $A$                                                                        |
| 5            | 5  | $a \in A$                                                                | $A$                                                                        |
| 6            | 6  | $X \in K_S$                                                              | $A$                                                                        |
| 1,4,6        | 7  | $(BX = z_S \rightarrow AX = z_S) \wedge (XB = z_S \rightarrow XA = z_S)$ | Def. 33.3.2.1(1, 4, 6)                                                     |
| 8            | 8  | $BX = z_S$                                                               | $A$                                                                        |
| 1,4,6,8      | 9  | $AX = z_S$                                                               | $\rightarrow E(\wedge E1(7), 8)$                                           |
| 1,2,4,5,6,8  | 10 | $\{a\}X = z_S$                                                           | Th. 33.3.1.6(3, 2, 6, 9, 5)                                                |
| 1,2,4,5,6    | 11 | $BX = z_S \rightarrow \{a\}X = z_S$                                      | $\rightarrow I(8, 10)$                                                     |
| 12           | 12 | $XB = z_S$                                                               | $A$                                                                        |
| 1,4,6,12     | 13 | $XA = z_S$                                                               | $\rightarrow E(\wedge E2(7), 12)$                                          |
| 1,2,4,5,6,12 | 14 | $X\{a\} = z_S$                                                           | Th. 33.3.1.7(3, 2, 6, 13, 5)                                               |
| 1,2,4,5,6    | 15 | $XB = z_S \rightarrow X\{a\} = z_S$                                      | $\rightarrow I(12, 14)$                                                    |
| 1,2,4,5      | 16 | $\{a\} \preceq_S B$                                                      | Def. 33.3.2.1(1, $\forall I$<br>( $\rightarrow I(6, \wedge I(11, 15))))$ ) |

|       |    |                                                               |                                   |
|-------|----|---------------------------------------------------------------|-----------------------------------|
| 1,2,4 | 17 | $\forall a \in A \{a\} \preceq_S B$                           | $\forall I(\rightarrow I(5, 16))$ |
| 1,2   | 18 | $A \preceq_S B \rightarrow \forall a \in A \{a\} \preceq_S B$ | $\rightarrow I(4, 17)$            |

Von den Elementen zur Menge:

|                   |    |                                                                                  |                                                                    |
|-------------------|----|----------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 19                | 19 | $\forall a \in A \{a\} \preceq_S B$                                              | $A$                                                                |
| 20                | 20 | $X \in K_S$                                                                      | $A$                                                                |
| 21                | 21 | $BX = z_S$                                                                       | $A$                                                                |
| 22                | 22 | $a \in A$                                                                        | $A$                                                                |
| 19,22             | 23 | $\{a\} \preceq_S B$                                                              | $\rightarrow E(\forall E(19), 22)$                                 |
| 1,19,20,<br>22    | 24 | $(BX = z_S \rightarrow \{a\}X = z_S) \wedge (XB = z_S \rightarrow X\{a\} = z_S)$ | Def. 33.3.2.1(1, 23, 20)                                           |
| 1,19,20,<br>21,22 | 25 | $\{a\}X = z_S$                                                                   | $\rightarrow E(\wedge E1(24), 21)$                                 |
| 1,19,20,<br>21    | 26 | $\forall a \in A \{a\}X = z_S$                                                   | $\forall I(\rightarrow I(22, 25))$                                 |
| 1,2,19,<br>20,21  | 27 | $AX = z_S$                                                                       | Th. 33.3.1.6(3, 2, 20, 26)                                         |
| 1,2,19,20         | 28 | $BX = z_S \rightarrow AX = z_S$                                                  | $\rightarrow I(21, 27)$                                            |
| 29                | 29 | $XB = z_S$                                                                       | $A$                                                                |
| 1,19,20,<br>22,29 | 30 | $X\{a\} = z_S$                                                                   | $\rightarrow E(\wedge E2(24), 29)$                                 |
| 1,19,20,<br>29    | 31 | $\forall a \in A X\{a\} = z_S$                                                   | $\forall I(\rightarrow I(22, 30))$                                 |
| 1,2,19,<br>20,29  | 32 | $XA = z_S$                                                                       | Th. 33.3.1.7(3, 2, 20, 31)                                         |
| 1,2,19,20         | 33 | $XB = z_S \rightarrow XA = z_S$                                                  | $\rightarrow I(29, 32)$                                            |
| 1,2,19            | 34 | $A \preceq_S B$                                                                  | Def. 33.3.2.1(1, $\forall I(\rightarrow I(20, \wedge I(28, 33))))$ |
| 1,2               | 35 | $(\forall a \in A \{a\} \preceq_S B) \rightarrow A \preceq_S B$                  | $\rightarrow I(19, 34)$                                            |
| 1,2               | 36 | $A \preceq_S B \leftrightarrow \forall a \in A \{a\} \preceq_S B$                | $\leftrightarrow I(18, 35)$                                        |

□

**Theorem 33.3.2.6** (Nulltests hängen nur von den beiden Profilen ab).

**Mengen:**  $S, A, A', B, B'$

$$H_S, A, A', B, B' \in K_S, A \sim_S A', B \sim_S B' \vdash AB = z_S \leftrightarrow A'B' = z_S$$

|             |    |                                        |                                             |
|-------------|----|----------------------------------------|---------------------------------------------|
| 1           | 1  | $H_S$                                  | $A$                                         |
| 2           | 2  | $A, A', B, B' \in K_S$                 | $A$                                         |
| 3           | 3  | $A \sim_S A'$                          | $A$                                         |
| 4           | 4  | $B \sim_S B'$                          | $A$                                         |
| 1,3         | 5  | $A \preceq_S A' \wedge A' \preceq_S A$ | Def. 33.3.2.2(1, 3)                         |
| 1,4         | 6  | $B \preceq_S B' \wedge B' \preceq_S B$ | Def. 33.3.2.2(1, 4)                         |
| 7           | 7  | $AB = z_S$                             | $A$                                         |
| 1,2,3,7     | 8  | $A'B = z_S$                            | Def. 33.3.2.1(1, $\wedge E2$<br>(5), 2, 7)  |
| 1,2,3,4,7   | 9  | $A'B' = z_S$                           | Def. 33.3.2.1(1, $\wedge E2$<br>(6), 2, 8)  |
| 1,2,3,4     | 10 | $AB = z_S \rightarrow A'B' = z_S$      | $\rightarrow I(7, 9)$                       |
| 11          | 11 | $A'B' = z_S$                           | $A$                                         |
| 1,2,3,11    | 12 | $AB' = z_S$                            | Def. 33.3.2.1(1, $\wedge E1$<br>(5), 2, 11) |
| 1,2,3,4, 11 | 13 | $AB = z_S$                             | Def. 33.3.2.1(1, $\wedge E1$<br>(6), 2, 12) |
| 1,2,3,4     | 14 | $A'B' = z_S \rightarrow AB = z_S$      | $\rightarrow I(11, 13)$                     |
| 1,2,3,4     | 15 | $AB = z_S \leftrightarrow A'B' = z_S$  | $\leftrightarrow I(10, 14)$                 |

**Theorem 33.3.2.7** (Das Hinzufügen der Null ändert ein Singleton-profil nicht).

**Mengen:**  $S, c, x, X$

$$H_S, c \in S \vdash \{c\} \sim_S \{0_S, c\}$$

*Beweis.*

Die beiden Nulltests:

|     |    |                                   |                                  |
|-----|----|-----------------------------------|----------------------------------|
| 1   | 1  | $H_S$                             | $A$                              |
| 2   | 2  | $c \in S$                         | $A$                              |
| 1   | 3  | $0_S \in S$                       | Ax. 33.2.1.2(1)                  |
| 1,2 | 4  | $\{0_S, c\} \subseteq S$          | Th. 3.13.3.74(3, 2)              |
|     | 5  | $c \in \{0_S, c\}$                | Th. 3.11.3.4                     |
|     | 6  | $\{0_S, c\} \neq \emptyset$       | Th. 3.6.3.5(5)                   |
| 1,2 | 7  | $\{0_S, c\} \in K_S$              | Th. 3.16.3.1( $\wedge I(4, 6)$ ) |
| 2   | 8  | $\{c\} \in K_S$                   | Th. 33.3.1.1(2)                  |
| 1   | 9  | $\text{HGr}(S, \star)$            | Ax. 33.2.1.1(1)                  |
| 10  | 10 | $X \in K_S$                       | $A$                              |
| 1   | 11 | $\text{HGr}_0(K_S, \star_P, z_S)$ | Th. 33.3.1.4(1)                  |

|                |    |                                              |                                                |
|----------------|----|----------------------------------------------|------------------------------------------------|
| 1,10           | 12 | $z_S X = z_S \wedge X z_S = z_S$             | Def. 33.2.1.1(11, 10)                          |
|                | 13 | $\{c\}X = z_S$                               | $A$                                            |
|                | 14 | $x \in \{0_S, c\}$                           | $A$                                            |
|                | 14 | $x = 0_S \vee x = c$                         | Th. 3.11.2.1(14)                               |
|                | 16 | $x = 0_S$                                    | $A$                                            |
| 1,10,16        | 17 | $\{x\}X = z_S$                               | $= E(\text{Th. 2.9.1.1 (16), } \wedge E1(12))$ |
|                | 18 | $x = c$                                      | $A$                                            |
| 13,18          | 19 | $\{x\}X = z_S$                               | $= E(\text{Th. 2.9.1.1 (18), 13})$             |
| 1,10,13,<br>14 | 20 | $\{x\}X = z_S$                               | $\vee E(15, [16]:17,$<br>$[18]:19)$            |
| 1,10,13        | 21 | $\forall x \in \{0_S, c\} \{x\}X = z_S$      | $\forall I(\rightarrow I(14, 20))$             |
| 1,2,10,13      | 22 | $\{0_S, c\}X = z_S$                          | Th. 33.3.1.6(9, 7, 10, 21)                     |
| 1,2,10         | 23 | $\{c\}X = z_S \rightarrow \{0_S, c\}X = z_S$ | $\rightarrow I(13, 22)$                        |
|                | 24 | $X\{c\} = z_S$                               | $A$                                            |
| 1,10,16        | 25 | $X\{x\} = z_S$                               | $= E(\text{Th. 2.9.1.1 (16), } \wedge E2(12))$ |
| 18,24          | 26 | $X\{x\} = z_S$                               | $= E(\text{Th. 2.9.1.1 (18), 24})$             |
| 1,10,14,<br>24 | 27 | $X\{x\} = z_S$                               | $\vee E(15, [16]:25,$<br>$[18]:26)$            |
| 1,10,24        | 28 | $\forall x \in \{0_S, c\} X\{x\} = z_S$      | $\forall I(\rightarrow I(14, 27))$             |
| 1,2,10,24      | 29 | $X\{0_S, c\} = z_S$                          | Th. 33.3.1.7(9, 7, 10, 28)                     |
| 1,2,10         | 30 | $X\{c\} = z_S \rightarrow X\{0_S, c\} = z_S$ | $\rightarrow I(24, 29)$                        |

Beide Profilrichtungen:

|           |    |                                              |                                                                               |
|-----------|----|----------------------------------------------|-------------------------------------------------------------------------------|
| 1,2       | 31 | $\{0_S, c\} \preceq_S \{c\}$                 | Def. 33.3.2.1(1, $\forall I$<br>$(\rightarrow I(10, \wedge I(23,$<br>$30))))$ |
|           | 32 | $\{0_S, c\}X = z_S$                          | $A$                                                                           |
| 1,2,10,32 | 33 | $\{c\}X = z_S$                               | Th. 33.3.1.6(9, 7, 10, 32, 5)                                                 |
| 1,2,10    | 34 | $\{0_S, c\}X = z_S \rightarrow \{c\}X = z_S$ | $\rightarrow I(32, 33)$                                                       |
|           | 35 | $X\{0_S, c\} = z_S$                          | $A$                                                                           |
| 1,2,10,35 | 36 | $X\{c\} = z_S$                               | Th. 33.3.1.7(9, 7, 10, 35, 5)                                                 |
| 1,2,10    | 37 | $X\{0_S, c\} = z_S \rightarrow X\{c\} = z_S$ | $\rightarrow I(35, 36)$                                                       |

|     |    |                              |                                                                              |
|-----|----|------------------------------|------------------------------------------------------------------------------|
| 1,2 | 38 | $\{c\} \preceq_S \{0_S, c\}$ | Def. 33.3.2.1(1, $\forall I$<br>( $\rightarrow I(10, \wedge I(34,$<br>37)))) |
| 1,2 | 39 | $\{c\} \sim_S \{0_S, c\}$    | Def. 33.3.2.2(1, $\wedge I$<br>(38, 31))                                     |

□

### 3.3 Untere Mengen und Grundelemente

**Theorem 33.3.3.1** (Vergleich mit einem beliebigen Profil).

*Mengen:*  $S, p, A, B, a$

$$H_S, p \in Q_S, A \in K_S \vdash ([A]_S \leq_S p \leftrightarrow \forall a \in A [\{a\}]_S \leq_S p)$$

|                |    |                                                                                          |                                                 |
|----------------|----|------------------------------------------------------------------------------------------|-------------------------------------------------|
| 1              | 1  | $H_S$                                                                                    | $A$                                             |
| 2              | 2  | $p \in Q_S$                                                                              | $A$                                             |
| 3              | 3  | $A \in K_S$                                                                              | $A$                                             |
| 1              | 4  | $\text{EqRel}(K_S, \sim_S)$                                                              | Th. 33.3.2.2(1)                                 |
| 1,2            | 5  | $\exists B \in K_S p = [B]_S$                                                            | Th. 11.4.2.4(4, 2)                              |
| 6              | 6  | $B \in K_S \wedge p = [B]_S$                                                             | $A$                                             |
| 1,3,6          | 7  | $[A]_S \leq_S [B]_S \leftrightarrow A \preceq_S B$                                       | Th. 33.3.2.4(1, 3, 6)                           |
| 1,3,6          | 8  | $A \preceq_S B \leftrightarrow \forall a \in A \{a\} \preceq_S B$                        | Th. 33.3.2.5(1, 3, 6)                           |
| 9              | 9  | $a \in A$                                                                                | $A$                                             |
| 3,9            | 10 | $\{a\} \in K_S$                                                                          | Th. 33.3.1.1(<br>Th. 28.2.4.2( $H1$ )(3,<br>9)) |
| 1,3,6,9        | 11 | $\{a\} \preceq_S B \leftrightarrow [\{a\}]_S \leq_S [B]_S$                               | Th. 2.1.2.5(<br>Th. 33.3.2.4(1, 10,<br>6))      |
| 1,3,6          | 12 | $\forall a \in A (\{a\} \preceq_S B \leftrightarrow [\{a\}]_S \leq_S [B]_S)$             | $\forall I(\rightarrow I(9, 11))$               |
| 13             | 13 | $\forall a \in A \{a\} \preceq_S B$                                                      | $A$                                             |
| 9,13           | 14 | $\{a\} \preceq_S B$                                                                      | $\rightarrow E(\forall E(13), 9)$               |
| 1,3,6,9,<br>13 | 15 | $[\{a\}]_S \leq_S [B]_S$                                                                 | Th. 2.2.4.1(11, 14)                             |
| 1,3,6,13       | 16 | $\forall a \in A [\{a\}]_S \leq_S [B]_S$                                                 | $\forall I(\rightarrow I(9, 15))$               |
| 1,3,6          | 17 | $(\forall a \in A \{a\} \preceq_S B) \rightarrow \forall a \in A [\{a\}]_S \leq_S [B]_S$ | $\rightarrow I(13, 16)$                         |
| 18             | 18 | $\forall a \in A [\{a\}]_S \leq_S [B]_S$                                                 | $A$                                             |
| 9,18           | 19 | $[\{a\}]_S \leq_S [B]_S$                                                                 | $\rightarrow E(\forall E(18), 9)$               |
| 1,3,6,9,<br>18 | 20 | $\{a\} \preceq_S B$                                                                      | Th. 2.2.4.2(11, 19)                             |
| 1,3,6,18       | 21 | $\forall a \in A \{a\} \preceq_S B$                                                      | $\forall I(\rightarrow I(9, 20))$               |

|       |    |                                                                                              |                                                   |
|-------|----|----------------------------------------------------------------------------------------------|---------------------------------------------------|
| 1,3,6 | 22 | $(\forall a \in A [\{a\}]_S \leq_S [B]_S) \rightarrow \forall a \in A \{a\} \preceq_S B$     | $\rightarrow I(18, 21)$                           |
| 1,3,6 | 23 | $(\forall a \in A \{a\} \preceq_S B) \leftrightarrow \forall a \in A [\{a\}]_S \leq_S [B]_S$ | $\leftrightarrow I(17, 22)$                       |
| 1,3,6 | 24 | $[A]_S \leq_S [B]_S \leftrightarrow \forall a \in A [\{a\}]_S \leq_S [B]_S$                  | Th. 2.2.3.5(<br>Th. 2.2.3.5(7, 8), 23)            |
| 1,3,6 | 25 | $[A]_S \leq_S p \leftrightarrow \forall a \in A [\{a\}]_S \leq_S p$                          | $= E(\text{Th. 2.9.1.1}$<br>$(\wedge E2(6)), 24)$ |
| 1,2,3 | 26 | $[A]_S \leq_S p \leftrightarrow \forall a \in A [\{a\}]_S \leq_S p$                          | $\exists E(5, 6, 25)$                             |

**Theorem 33.3.3.2 (Untere Mengen sind nichtleere Potenzmengen).**

**Mengen:**  $S, p, A, a, L_p, D_p$

$$H_S, p \in Q_S \vdash L_p = \mathcal{P}(D_p) \setminus \{\emptyset\}$$

*Beweis.*

Von der unteren Menge zur Potenzmenge:

|         |    |                                                          |                                                      |
|---------|----|----------------------------------------------------------|------------------------------------------------------|
| 1       | 1  | $H_S$                                                    | $A$                                                  |
|         | 2  | $p \in Q_S$                                              | $A$                                                  |
|         | 3  | $A \in L_p$                                              | $A$                                                  |
| 1,3     | 4  | $A \in K_S \wedge [A]_S \leq_S p$                        | Def. 33.3.2.3(1, 3)                                  |
| 1,2,3   | 5  | $\forall a \in A [\{a\}]_S \leq_S p$                     | Th. 33.3.3.1(1, 2,<br>$\wedge E1(4), \wedge E2(4)$ ) |
|         | 6  | $a \in A$                                                | $A$                                                  |
| 1,3,6   | 7  | $a \in S$                                                | Th. 28.2.4.2( $H1$ )<br>$(\wedge E1(4), 6)$          |
| 1,2,3,6 | 8  | $[\{a\}]_S \leq_S p$                                     | $\rightarrow E(\forall E(5), 6)$                     |
| 1,2,3,6 | 9  | $a \in D_p$                                              | Def. 33.3.2.3(1, $\wedge I$<br>(7, 8))               |
| 1,2,3   | 10 | $A \subseteq D_p$                                        | Th. 3.5.1.1( $[6] \dot{:} 9$ )                       |
| 1,3     | 11 | $A \neq \emptyset$                                       | $\wedge E2(\text{Th. 3.16.3.1}$<br>$(\wedge E1(4)))$ |
| 1,2,3   | 12 | $A \in \mathcal{P}(D_p) \setminus \{\emptyset\}$         | Th. 3.16.3.1( $\wedge I(10,$<br>$11)$ )              |
| 1,2     | 13 | $L_p \subseteq \mathcal{P}(D_p) \setminus \{\emptyset\}$ | Th. 3.5.1.1( $[3] \dot{:} 12$ )                      |

Von der Potenzmenge zur unteren Menge:

|    |    |                                                  |                                   |
|----|----|--------------------------------------------------|-----------------------------------|
| 14 | 14 | $A \in \mathcal{P}(D_p) \setminus \{\emptyset\}$ | $A$                               |
| 14 | 15 | $A \subseteq D_p \wedge A \neq \emptyset$        | Th. 3.16.3.1(14)                  |
| 1  | 16 | $D_p \subseteq S$                                | Th. 3.7.3.7(<br>Def. 33.3.2.3(1)) |

|         |    |                                                          |                                                    |
|---------|----|----------------------------------------------------------|----------------------------------------------------|
| 1,14    | 17 | $A \subseteq S$                                          | Th. 3.5.3.6( $\wedge E1$<br>(15), 16)              |
| 1,14    | 18 | $A \in K_S$                                              | Th. 3.16.3.1( $\wedge I(17,$<br>$\wedge E2(15))$ ) |
| 19      | 19 | $a \in A$                                                | $A$                                                |
| 14,19   | 20 | $a \in D_p$                                              | Th. 3.5.2.6( $\wedge E1$<br>(15), 19)              |
| 1,14,19 | 21 | $[\{a\}]_S \leq_S p$                                     | $\wedge E2$ (Def. 33.3.2.3<br>(1, 20))             |
| 1,14    | 22 | $\forall a \in A [\{a\}]_S \leq_S p$                     | $\forall I(\rightarrow I(19, 21))$                 |
| 1,2,14  | 23 | $[A]_S \leq_S p$                                         | Th. 33.3.3.1(1, 2, 18,<br>22)                      |
| 1,2,14  | 24 | $A \in L_p$                                              | Def. 33.3.2.3(1, $\wedge I$<br>(18, 23))           |
| 1,2     | 25 | $\mathcal{P}(D_p) \setminus \{\emptyset\} \subseteq L_p$ | Th. 3.5.1.1( $[14]:24$ )                           |
| 1,2     | 26 | $L_p = \mathcal{P}(D_p) \setminus \{\emptyset\}$         | Th. 3.5.3.5(13, 25)                                |

□

### 3.4 Transport durch Isomorphismen

**Theorem 33.3.4.1 (Transport der Nullproduktrelation).**

**Mengen:**  $S, T, A, B$

$$H_S, H_T, I, A, B \in K_S \vdash (AB = z_S \leftrightarrow \Phi(A)\Phi(B) = z_T)$$

|         |    |                                                 |                                    |
|---------|----|-------------------------------------------------|------------------------------------|
| 1       | 1  | $H_S$                                           | $A$                                |
| 2       | 2  | $H_T$                                           | $A$                                |
| 3       | 3  | $I$                                             | $A$                                |
| 4       | 4  | $A, B \in K_S$                                  | $A$                                |
| 1,2,3   | 5  | $\Phi(z_S) = z_T$                               | Th. 33.3.1.5(1, 2, 3)              |
| 3       | 6  | $\text{HGr}(K_S, \star \mathcal{P})$            | Th. 28.2.15.2( $H1$ )<br>(3)       |
| 3,4     | 7  | $AB \in K_S$                                    | Ax. 28.2.1.1(6, 4)                 |
| 1       | 8  | $z_S \in K_S$                                   | Ax. 33.2.1.2(<br>Th. 33.3.1.4(1))  |
| 3,4     | 9  | $\Phi(AB) = \Phi(A)\Phi(B)$                     | Th. 28.2.15.2( $H5$ )<br>(3, 4)    |
| 1,3,4   | 10 | $AB = z_S \leftrightarrow \Phi(AB) = \Phi(z_S)$ | Th. 28.2.15.2( $H6$ )<br>(3, 7, 8) |
| 1,2,3,4 | 11 | $AB = z_S \leftrightarrow \Phi(A)\Phi(B) = z_T$ | $= E(5, = E(9, 10))$               |

**Theorem 33.3.4.2 (Isomorphismen erhalten den Profilvergleich).**

**Mengen:**  $S, T, A, B, X, Y$

$$H_S, H_T, I, A, B \in K_S, A \preceq_S B \vdash \Phi(A) \preceq_T \Phi(B)$$

*Beweis.*

Urbild eines beliebigen Nulltests:

|         |    |                                                                                                                           |                        |
|---------|----|---------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1       | 1  | $H_S$                                                                                                                     | $A$                    |
| 2       | 2  | $H_T$                                                                                                                     | $A$                    |
| 3       | 3  | $I$                                                                                                                       | $A$                    |
| 4       | 4  | $A, B \in K_S$                                                                                                            | $A$                    |
| 5       | 5  | $A \preceq_S B$                                                                                                           | $A$                    |
| 6       | 6  | $Y \in K_T$                                                                                                               | $A$                    |
| 3       | 7  | $\Phi : K_S \xrightarrow{\sim} K_T$                                                                                       | Ax. 28.2.12.6(3)       |
| 3       | 8  | $\Phi^{-1} : K_T \rightarrow K_S$                                                                                         | Th. 8.3.3.1(7)         |
| 3,6     | 9  | $\Phi^{-1}(Y) \in K_S$                                                                                                    | Th. 5.2.3.9(8, 6)      |
| 3,6     | 10 | $\Phi(\Phi^{-1}(Y)) = Y$                                                                                                  | Th. 8.3.3.6(7, 6)      |
| 1,3,5,6 | 11 | $(B\Phi^{-1}(Y) = z_S \rightarrow A\Phi^{-1}(Y) = z_S)$<br>$\wedge (\Phi^{-1}(Y)B = z_S \rightarrow \Phi^{-1}(Y)A = z_S)$ | Def. 33.3.2.1(1, 5, 9) |

Rechter und linker Test:

|                    |    |                                             |                                                                       |
|--------------------|----|---------------------------------------------|-----------------------------------------------------------------------|
| 12                 | 12 | $\Phi(B)Y = z_T$                            | $A$                                                                   |
| 1,2,3,4,6,<br>12   | 13 | $B\Phi^{-1}(Y) = z_S$                       | Th. 33.3.4.1(1, 2, 3,<br>4, 9, = $E(\text{Th. 2.9.1.1}$<br>(10), 12)) |
| 1,2,3,4,5,<br>6,12 | 14 | $A\Phi^{-1}(Y) = z_S$                       | $\rightarrow E(\wedge E1(11), 13)$                                    |
| 1,2,3,4,5,<br>6,12 | 15 | $\Phi(A)Y = z_T$                            | $= E(10, \text{Th. 33.3.4.1}$<br>(1, 2, 3, 4, 9, 14))                 |
| 1,2,3,4,5,<br>6    | 16 | $\Phi(B)Y = z_T \rightarrow \Phi(A)Y = z_T$ | $\rightarrow I(12, 15)$                                               |
| 17                 | 17 | $Y\Phi(B) = z_T$                            | $A$                                                                   |
| 1,2,3,4,6,<br>17   | 18 | $\Phi^{-1}(Y)B = z_S$                       | Th. 33.3.4.1(1, 2, 3,<br>9, 4, = $E(\text{Th. 2.9.1.1}$<br>(10), 17)) |
| 1,2,3,4,5,<br>6,17 | 19 | $\Phi^{-1}(Y)A = z_S$                       | $\rightarrow E(\wedge E2(11), 18)$                                    |
| 1,2,3,4,5,<br>6,17 | 20 | $Y\Phi(A) = z_T$                            | $= E(10, \text{Th. 33.3.4.1}$<br>(1, 2, 3, 9, 4, 19))                 |
| 1,2,3,4,5,<br>6    | 21 | $Y\Phi(B) = z_T \rightarrow Y\Phi(A) = z_T$ | $\rightarrow I(17, 20)$                                               |

1,2,3,4,5 22  $\Phi(A) \preceq_T \Phi(B)$

Def. 33.3.2.1(2,  $\forall I$   
( $\rightarrow I(6, \wedge I(16,$   
21))))

□

**Theorem 33.3.4.3 (Isomorphismen erhalten und reflektieren Profilvergleiche).**

**Mengen:**  $S, T, A, B$

$$H_S, H_T, I, A, B \in K_S \vdash (A \preceq_S B \leftrightarrow \Phi(A) \preceq_T \Phi(B))$$

|             |    |                                                           |                                         |
|-------------|----|-----------------------------------------------------------|-----------------------------------------|
| 1           | 1  | $H_S$                                                     | $A$                                     |
| 2           | 2  | $H_T$                                                     | $A$                                     |
| 3           | 3  | $I$                                                       | $A$                                     |
| 4           | 4  | $A, B \in K_S$                                            | $A$                                     |
| 5           | 5  | $A \preceq_S B$                                           | $A$                                     |
| 1,2,3,4,5   | 6  | $\Phi(A) \preceq_T \Phi(B)$                               | Th. 33.3.4.2(1, 2, 3, 4, 5)             |
| 1,2,3,4     | 7  | $A \preceq_S B \rightarrow \Phi(A) \preceq_T \Phi(B)$     | $\rightarrow I(5, 6)$                   |
| 3           | 8  | $\text{HGrIso}(\Phi^{-1}; K_T, \diamond_P; K_S, \star_P)$ | Th. 28.2.15.1(3)                        |
| 3           | 9  | $\Phi : K_S \xrightarrow{\sim} K_T$                       | Ax. 28.2.12.6(3)                        |
| 3,4         | 10 | $\Phi(A), \Phi(B) \in K_T$                                | Th. 5.2.3.9(<br>Def. 8.2.1.1(9), 4)     |
| 11          | 11 | $\Phi(A) \preceq_T \Phi(B)$                               | $A$                                     |
| 1,2,3,4, 11 | 12 | $\Phi^{-1}(\Phi(A)) \preceq_S \Phi^{-1}(\Phi(B))$         | Th. 33.3.4.2(2, 1, 8, 10, 11)           |
| 3,4         | 13 | $\Phi^{-1}(\Phi(A)) = A \wedge \Phi^{-1}(\Phi(B)) = B$    | Th. 8.3.3.5(9, 4)                       |
| 1,2,3,4, 11 | 14 | $A \preceq_S B$                                           | $= E(\wedge E1(13), \wedge E2(13), 12)$ |
| 1,2,3,4     | 15 | $\Phi(A) \preceq_T \Phi(B) \rightarrow A \preceq_S B$     | $\rightarrow I(11, 14)$                 |
| 1,2,3,4     | 16 | $A \preceq_S B \leftrightarrow \Phi(A) \preceq_T \Phi(B)$ | $\leftrightarrow I(7, 15)$              |

**Theorem 33.3.4.4 (Transport der Profiläquivalenz).**

**Mengen:**  $S, T, A, B$

$$H_S, H_T, I, A, B \in K_S \vdash (A \sim_S B \leftrightarrow \Phi(A) \sim_T \Phi(B))$$

|   |   |       |     |
|---|---|-------|-----|
| 1 | 1 | $H_S$ | $A$ |
|---|---|-------|-----|

|         |    |                                                                                                       |                                                 |
|---------|----|-------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 2       | 2  | $H_T$                                                                                                 | $A$                                             |
| 3       | 3  | $I$                                                                                                   | $A$                                             |
| 4       | 4  | $A, B \in K_S$                                                                                        | $A$                                             |
| 1,2,3,4 | 5  | $A \preceq_S B \leftrightarrow \Phi(A) \preceq_T \Phi(B)$                                             | Th. 33.3.4.3(1, 2, 3, 4)                        |
| 1,2,3,4 | 6  | $B \preceq_S A \leftrightarrow \Phi(B) \preceq_T \Phi(A)$                                             | Th. 33.3.4.3(1, 2, 3, 4)                        |
| 1       | 7  | $A \sim_S B \leftrightarrow (A \preceq_S B \wedge B \preceq_S A)$                                     | Def. 33.3.2.2(1)                                |
| 2       | 8  | $\Phi(A) \sim_T \Phi(B) \leftrightarrow (\Phi(A) \preceq_T \Phi(B) \wedge \Phi(B) \preceq_T \Phi(A))$ | Def. 33.3.2.2(2)                                |
| 1,2,3,4 | 9  | $A \sim_S B \leftrightarrow (\Phi(A) \preceq_T \Phi(B) \wedge \Phi(B) \preceq_T \Phi(A))$             | $\leftrightarrow S(6, \leftrightarrow S(5, 7))$ |
| 1,2,3,4 | 10 | $A \sim_S B \leftrightarrow \Phi(A) \sim_T \Phi(B)$                                                   | Th. 2.2.3.6(9, 8)                               |

Zur Konstruktion der Profilabbildung seien  $\pi_S = q_{K_S, \sim_S}$  und  $\pi_T = q_{K_T, \sim_T}$ . Die folgende Konstruktion whlt keine Reprsentanten aus den Profilklassen.

**Theorem 33.3.4.5 (Existenz der induzierten Profilabbildung).**

**Mengen:**  $S, T, A, B$

$$H_S, H_T, I \vdash \exists! g (g : Q_S \rightarrow Q_T \wedge g \circ \pi_S = \pi_T \circ \Phi)$$

|                |    |                                                                                        |                              |
|----------------|----|----------------------------------------------------------------------------------------|------------------------------|
| 1              | 1  | $H_S$                                                                                  | $A$                          |
| 2              | 2  | $H_T$                                                                                  | $A$                          |
| 3              | 3  | $I$                                                                                    | $A$                          |
| 1              | 4  | $\text{EqRel}(K_S, \sim_S)$                                                            | Th. 33.3.2.2(1)              |
| 2              | 5  | $\text{EqRel}(K_T, \sim_T)$                                                            | Th. 33.3.2.2(2)              |
| 2              | 6  | $\pi_T : K_T \rightarrow Q_T$                                                          | Th. 11.5.1.1(5)              |
| 3              | 7  | $\Phi : K_S \rightarrow K_T$                                                           | Th. 28.2.15.2(H3)<br>(3)     |
| 2,3            | 8  | $\pi_T \circ \Phi : K_S \rightarrow Q_T$                                               | Def. 5.3.3.1(7, 6)           |
| 9              | 9  | $A, B \in K_S$                                                                         | $A$                          |
| 10             | 10 | $A \sim_S B$                                                                           | $A$                          |
| 1,2,3,9,<br>10 | 11 | $\Phi(A) \sim_T \Phi(B)$                                                               | Th. 33.3.4.4(1, 2, 3, 9, 10) |
| 3,9            | 12 | $\Phi(A), \Phi(B) \in K_T$                                                             | Th. 5.2.3.9(7, 9)            |
| 1,2,3,9,<br>10 | 13 | $[\Phi(A)]_T = [\Phi(B)]_T$                                                            | Th. 11.4.1.5(5, 12, 11)      |
| 2,3,9          | 14 | $\pi_T(\Phi(A)) = [\Phi(A)]_T \wedge \pi_T(\Phi(B)) = [\Phi(B)]_T$                     | Th. 11.5.1.2(5, 12)          |
| 2,3,9          | 15 | $(\pi_T \circ \Phi)(A) = \pi_T(\Phi(A)) \wedge (\pi_T \circ \Phi)(B) = \pi_T(\Phi(B))$ | Def. 5.3.3.1(7, 6, 9)        |

|                |    |                                                                                                    |                                                                                                                                                                                         |
|----------------|----|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1,2,3,9,<br>10 | 16 | $(\pi_T \circ \Phi)(A) = (\pi_T \circ \Phi)(B)$                                                    | $= E(\text{Th. 2.9.1.1}$<br>$(\wedge E1(14)),$<br>$\text{Th. 2.9.1.1}(\wedge E2$<br>$(14)), \text{Th. 2.9.1.1}$<br>$(\wedge E1(15)),$<br>$\text{Th. 2.9.1.1}(\wedge E2$<br>$(15)), 13)$ |
| 1,2,3          | 17 | $\forall A, B \in K_S (A \sim_S B \rightarrow (\pi_T \circ \Phi)(A) =$<br>$(\pi_T \circ \Phi)(B))$ | $\forall I(\forall I(\rightarrow I(9, \rightarrow I$<br>$(10, 16))))$                                                                                                                   |
| 1,2,3          | 18 | $\exists! g (g : Q_S \rightarrow Q_T \wedge g \circ \pi_S = \pi_T \circ \Phi)$                     | $\text{Th. 11.5.2.2}(4, 8, 17)$                                                                                                                                                         |

**Definition 33.3.4.1 (Die induzierte Profllabbildung).**

*Quellhalbgruppe mit Null:*  $\triangleright H_S$   
*Zielhalbgruppe mit Null:*  $\triangleright H_T$   
*Potenzisomorphismus:*  $\triangleright I$   
*Eindeutigkeit:*  $\text{Th. 33.3.4.5}$

$$\psi := \iota g (g : Q_S \rightarrow Q_T \wedge g \circ \pi_S = \pi_T \circ \Phi)$$

**Theorem 33.3.4.6 (Typ und Repräsentantenwert der Profllabbildung).**

*Mengen:*  $S, T, A$

$$H_S, H_T, I \vdash \psi : Q_S \rightarrow Q_T \wedge \forall A \in K_S (\psi([A]_S) = [\Phi(A)]_T)$$

|       |    |                                                                         |                                                      |
|-------|----|-------------------------------------------------------------------------|------------------------------------------------------|
| 1     | 1  | $H_S$                                                                   | $A$                                                  |
| 2     | 2  | $H_T$                                                                   | $A$                                                  |
| 3     | 3  | $I$                                                                     | $A$                                                  |
| 1,2,3 | 4  | $\psi : Q_S \rightarrow Q_T \wedge \psi \circ \pi_S = \pi_T \circ \Phi$ | $\text{Def. 33.3.4.1}(\text{Th. 33.3.4.5}(1, 2, 3))$ |
| 5     | 5  | $A \in K_S$                                                             | $A$                                                  |
| 1     | 6  | $\text{EqRel}(K_S, \sim_S)$                                             | $\text{Th. 33.3.2.2}(1)$                             |
| 2     | 7  | $\text{EqRel}(K_T, \sim_T)$                                             | $\text{Th. 33.3.2.2}(2)$                             |
| 3     | 8  | $\Phi : K_S \rightarrow K_T$                                            | $\text{Th. 28.2.15.2}(H3)$<br>$(3)$                  |
| 3,5   | 9  | $\Phi(A) \in K_T$                                                       | $\text{Th. 5.2.3.9}(8, 5)$                           |
| 1     | 10 | $\pi_S : K_S \rightarrow Q_S$                                           | $\text{Th. 11.5.1.1}(6)$                             |
| 2     | 11 | $\pi_T : K_T \rightarrow Q_T$                                           | $\text{Th. 11.5.1.1}(7)$                             |
| 1,5   | 12 | $\pi_S(A) = [A]_S$                                                      | $\text{Th. 11.5.1.2}(6, 5)$                          |
| 2,3,5 | 13 | $\pi_T(\Phi(A)) = [\Phi(A)]_T$                                          | $\text{Th. 11.5.1.2}(7, 9)$                          |

|         |    |                                                                                   |                                                             |
|---------|----|-----------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1,2,3,5 | 14 | $(\psi \circ \pi_S)(A) = \psi(\pi_S(A))$                                          | Def. 5.3.3.1(10, $\wedge E1$<br>(4), 5)                     |
| 2,3,5   | 15 | $(\pi_T \circ \Phi)(A) = \pi_T(\Phi(A))$                                          | Def. 5.3.3.1(8, 11, 5)                                      |
| 1,2,3,5 | 16 | $\psi([A]_S) = [\Phi(A)]_T$                                                       | Th. 2.9.1.1(= $E$<br>( $\wedge E2$ (4), 15, 12, 13,<br>14)) |
| 1,2,3   | 17 | $\forall A \in K_S (\psi([A]_S) = [\Phi(A)]_T)$                                   | $\forall I(\rightarrow I(5, 16))$                           |
| 1,2,3   | 18 | $\psi : Q_S \rightarrow Q_T \wedge \forall A \in K_S (\psi([A]_S) = [\Phi(A)]_T)$ | $\wedge I(\wedge E1(4), 17)$                                |

**Theorem 33.3.4.7 (Die Profilabbildung erhält und reflektiert die Ordnung).**

**Mengen:**  $S, T, p, q, A, B$

$$H_S, H_T, I, p, q \in Q_S \vdash (p \leq_S q \leftrightarrow \psi(p) \leq_T \psi(q))$$

|           |    |                                                                            |                                                                                                                               |
|-----------|----|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 1         | 1  | $H_S$                                                                      | $A$                                                                                                                           |
| 2         | 2  | $H_T$                                                                      | $A$                                                                                                                           |
| 3         | 3  | $I$                                                                        | $A$                                                                                                                           |
| 4         | 4  | $p, q \in Q_S$                                                             | $A$                                                                                                                           |
| 1         | 5  | $\text{EqRel}(K_S, \sim_S)$                                                | Th. 33.3.2.2(1)                                                                                                               |
| 1,4       | 6  | $\exists A \in K_S p = [A]_S$                                              | Th. 11.4.2.4(5, 4)                                                                                                            |
| 1,4       | 7  | $\exists B \in K_S q = [B]_S$                                              | Th. 11.4.2.4(5, 4)                                                                                                            |
| 8         | 8  | $A \in K_S \wedge p = [A]_S$                                               | $A$                                                                                                                           |
| 9         | 9  | $B \in K_S \wedge q = [B]_S$                                               | $A$                                                                                                                           |
| 1,8,9     | 10 | $[A]_S \leq_S [B]_S \leftrightarrow A \preceq_S B$                         | Th. 33.3.2.4(1, 8, 9)                                                                                                         |
| 1,2,3,8,9 | 11 | $A \preceq_S B \leftrightarrow \Phi(A) \preceq_T \Phi(B)$                  | Th. 33.3.4.3(1, 2, 3,<br>8, 9)                                                                                                |
| 3         | 12 | $\Phi : K_S \rightarrow K_T$                                               | Th. 28.2.15.2( $H3$ )<br>(3)                                                                                                  |
| 3,8,9     | 13 | $\Phi(A), \Phi(B) \in K_T$                                                 | Th. 5.2.3.9(12, 8, 9)                                                                                                         |
| 2,3,8,9   | 14 | $[\Phi(A)]_T \leq_T [\Phi(B)]_T \leftrightarrow \Phi(A) \preceq_T \Phi(B)$ | Th. 33.3.2.4(2, 13)                                                                                                           |
| 1,2,3,8,9 | 15 | $\psi([A]_S) = [\Phi(A)]_T \wedge \psi([B]_S) = [\Phi(B)]_T$               | Th. 33.3.4.6(1, 2, 3,<br>8, 9)                                                                                                |
| 1,2,3,8,9 | 16 | $[A]_S \leq_S [B]_S \leftrightarrow \psi([A]_S) \leq_T \psi([B]_S)$        | = $E$ (Th. 2.9.1.1<br>( $\wedge E1$ (15)),<br>Th. 2.9.1.1( $\wedge E2$<br>(15)), Th. 2.2.3.6(<br>Th. 2.2.3.5(10, 11),<br>14)) |
| 1,2,3,8,9 | 17 | $p \leq_S q \leftrightarrow \psi(p) \leq_T \psi(q)$                        | = $E$ (Th. 2.9.1.1<br>( $\wedge E2$ (8)), Th. 2.9.1.1<br>( $\wedge E2$ (9)), 16)                                              |

$$1,2,3,4 \quad 18 \quad p \leq_S q \leftrightarrow \psi(p) \leq_T \psi(q) \qquad \exists E(6, [8]; \exists E(7, [9]; 17))$$

**Theorem 33.3.4.8 (Die Profilabbildung ist bijektiv).**

**Mengen:**  $S, T, p, q, A, B$

$$H_S, H_T, I \vdash \psi : Q_S \xrightarrow{\sim} Q_T$$

*Beweis.*

Injektivität aus der reflektierten Ordnung:

|           |    |                                                        |                                                                                         |
|-----------|----|--------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 1         | 1  | $H_S$                                                  | $A$                                                                                     |
| 2         | 2  | $H_T$                                                  | $A$                                                                                     |
| 3         | 3  | $I$                                                    | $A$                                                                                     |
| 1,2,3     | 4  | $\psi : Q_S \rightarrow Q_T$                           | $\wedge E1(\text{Th. 33.3.4.6}(1, 2, 3))$                                               |
| 1         | 5  | $\text{PartOrd}(Q_S, \leq_S)$                          | $\text{Th. 33.3.2.3}(1)$                                                                |
| 2         | 6  | $\text{PartOrd}(Q_T, \leq_T)$                          | $\text{Th. 33.3.2.3}(2)$                                                                |
| 7         | 7  | $p, q \in Q_S$                                         | $A$                                                                                     |
| 8         | 8  | $\psi(p) = \psi(q)$                                    | $A$                                                                                     |
| 1,2,3,7   | 9  | $\psi(p), \psi(q) \in Q_T$                             | $\text{Th. 5.2.3.9}(4, 7)$                                                              |
| 1,2,3,7   | 10 | $\psi(p) \leq_T \psi(p)$                               | $\text{Def. 12.2.1.1}(6, 9)$                                                            |
| 1,2,3,7,8 | 11 | $\psi(p) \leq_T \psi(q) \wedge \psi(q) \leq_T \psi(p)$ | $\wedge I(= E(8, 10), = E(8, 10))$                                                      |
| 1,2,3,7,8 | 12 | $p \leq_S q \wedge q \leq_S p$                         | $\text{Th. 33.3.4.7}(1, 2, 3, 7, 11)$                                                   |
| 1,2,3,7,8 | 13 | $p = q$                                                | $\text{Ax. 12.2.1.1}(5, 7, 12)$                                                         |
| 1,2,3     | 14 | $\psi : Q_S \rightarrowtail Q_T$                       | $\text{Def. 6.2.1.1}(4, \forall I(\forall I(\rightarrow I(7, \rightarrow I(8, 13))))))$ |

Surjektivität aus Repräsentanten:

|      |    |                                     |                               |
|------|----|-------------------------------------|-------------------------------|
| 15   | 15 | $q \in Q_T$                         | $A$                           |
| 2    | 16 | $\text{EqRel}(K_T, \sim_T)$         | $\text{Th. 33.3.2.2}(2)$      |
| 2,15 | 17 | $\exists B \in K_T \quad q = [B]_T$ | $\text{Th. 11.4.2.4}(16, 15)$ |
| 18   | 18 | $B \in K_T \wedge q = [B]_T$        | $A$                           |
| 3    | 19 | $\Phi : K_S \xrightarrow{\sim} K_T$ | $\text{Ax. 28.2.12.6}(3)$     |
| 3    | 20 | $\Phi^{-1} : K_T \rightarrow K_S$   | $\text{Th. 8.3.3.1}(19)$      |
| 3,18 | 21 | $\Phi^{-1}(B) \in K_S$              | $\text{Th. 5.2.3.9}(20, 18)$  |
| 1    | 22 | $\text{EqRel}(K_S, \sim_S)$         | $\text{Th. 33.3.2.2}(1)$      |

|          |    |                                                   |                                                      |
|----------|----|---------------------------------------------------|------------------------------------------------------|
| 1,3,18   | 23 | $[\Phi^{-1}(B)]_S \in Q_S$                        | Th. 11.4.2.2(22, 21)                                 |
| 1,2,3,18 | 24 | $\psi([\Phi^{-1}(B)]_S) = [\Phi(\Phi^{-1}(B))]_T$ | Th. 33.3.4.6(1, 2, 3, 21)                            |
| 3,18     | 25 | $\Phi(\Phi^{-1}(B)) = B$                          | Th. 8.3.3.6(19, 18)                                  |
| 1,2,3,18 | 26 | $\psi([\Phi^{-1}(B)]_S) = q$                      | $= E(25, \text{Th. 2.9.1.1}(\wedge E2(18)), 24)$     |
| 1,2,3,18 | 27 | $\exists p \in Q_S \psi(p) = q$                   | $\exists I(\wedge I(23, 26))$                        |
| 1,2,3,15 | 28 | $\exists p \in Q_S \psi(p) = q$                   | $\exists E(17, [18]:27)$                             |
| 1,2,3    | 29 | $\psi : Q_S \rightarrow Q_T$                      | Def. 7.2.1.1(4, $\forall I(\rightarrow I(15, 28))$ ) |
| 1,2,3    | 30 | $\psi : Q_S \xrightarrow{\sim} Q_T$               | Def. 8.2.1.1(14, 29)                                 |

□

**Theorem 33.3.4.9 (Bijektiver Transport der unteren Mengen).**

**Mengen:**  $S, T, p, A, L_p, L'_{\psi(p)}$

$$H_S, H_T, I, p \in Q_S \vdash \Phi \upharpoonright_{L_p} : L_p \xrightarrow{\sim} L'_{\psi(p)}$$

|           |    |                                                                       |                                                  |
|-----------|----|-----------------------------------------------------------------------|--------------------------------------------------|
| 1         | 1  | $H_S$                                                                 | $A$                                              |
| 2         | 2  | $H_T$                                                                 | $A$                                              |
| 3         | 3  | $I$                                                                   | $A$                                              |
| 4         | 4  | $p \in Q_S$                                                           | $A$                                              |
| 3         | 5  | $\Phi : K_S \xrightarrow{\sim} K_T$                                   | Ax. 28.2.12.6(3)                                 |
| 1         | 6  | $L_p \subseteq K_S$                                                   | Th. 3.7.3.7(Def. 33.3.2.3(1))                    |
| 2         | 7  | $L'_{\psi(p)} \subseteq K_T$                                          | Th. 3.7.3.7(Def. 33.3.2.3(2))                    |
| 8         | 8  | $A \in K_S$                                                           | $A$                                              |
| 3,8       | 9  | $\Phi(A) \in K_T$                                                     | Th. 5.2.3.9(Def. 8.2.1.1(5), 8)                  |
| 1         | 10 | $\text{EqRel}(K_S, \sim_S)$                                           | Th. 33.3.2.2(1)                                  |
| 1,8       | 11 | $[A]_S \in Q_S$                                                       | Th. 11.4.2.2(10, 8)                              |
| 1,2,3,8   | 12 | $\psi([A]_S) = [\Phi(A)]_T$                                           | Th. 33.3.4.6(1, 2, 3, 8)                         |
| 1,2,3,4,8 | 13 | $[A]_S \leq_S p \leftrightarrow \psi([A]_S) \leq_T \psi(p)$           | Th. 33.3.4.7(1, 2, 3, 11, 4)                     |
| 1,8       | 14 | $A \in L_p \leftrightarrow [A]_S \leq_S p$                            | Def. 33.3.2.3(1, 8)                              |
| 2,3,8     | 15 | $\Phi(A) \in L'_{\psi(p)} \leftrightarrow [\Phi(A)]_T \leq_T \psi(p)$ | Def. 33.3.2.3(2, 9)                              |
| 1,2,3,4,8 | 16 | $A \in L_p \leftrightarrow \Phi(A) \in L'_{\psi(p)}$                  | Th. 2.2.3.6(Th. 2.2.3.5(14, = $E(12, 13)$ ), 15) |

|         |    |                                                                            |                                     |
|---------|----|----------------------------------------------------------------------------|-------------------------------------|
| 1,2,3,4 | 17 | $\forall A \in K_S \ (A \in L_p \leftrightarrow \Phi(A) \in L'_{\psi(p)})$ | $\forall I(\rightarrow I(8,16))$    |
| 1,2,3,4 | 18 | $\Phi \upharpoonright_{L_p} : L_p \xrightarrow{\sim} L'_{\psi(p)}$         | Th. 8.3.2.2( $H4$ )(5,<br>6, 7, 17) |

**Theorem 33.3.4.10 (Gleich große untere Grundelementmengen).**

**Mengen:**  $S, T, p, D_p, D'_{\psi(p)}$

$$H_S, H_T, I, \text{Finite}(S), \text{Finite}(T), p \in Q_S \vdash \#(D_p) = \#(D'_{\psi(p)})$$

|                 |    |                                                                                                      |                                        |
|-----------------|----|------------------------------------------------------------------------------------------------------|----------------------------------------|
| 1               | 1  | $H_S$                                                                                                | $A$                                    |
| 2               | 2  | $H_T$                                                                                                | $A$                                    |
| 3               | 3  | $I$                                                                                                  | $A$                                    |
| 4               | 4  | $\text{Finite}(S)$                                                                                   | $A$                                    |
| 5               | 5  | $\text{Finite}(T)$                                                                                   | $A$                                    |
| 6               | 6  | $p \in Q_S$                                                                                          | $A$                                    |
| 1               | 7  | $D_p \subseteq S$                                                                                    | Th. 3.7.3.7(<br>Def. 33.3.2.3(1))      |
| 2               | 8  | $D'_{\psi(p)} \subseteq T$                                                                           | Th. 3.7.3.7(<br>Def. 33.3.2.3(2))      |
| 1,4             | 9  | $\text{Finite}(D_p)$                                                                                 | Th. 20.3.6.10(7, 4)                    |
| 2,5             | 10 | $\text{Finite}(D'_{\psi(p)})$                                                                        | Th. 20.3.6.10(8, 5)                    |
| 1,2,3,6         | 11 | $\Phi \upharpoonright_{L_p} : L_p \xrightarrow{\sim} L'_{\psi(p)}$                                   | Th. 33.3.4.9(1, 2, 3,<br>6)            |
| 1,2,3,6         | 12 | $L_p \approx L'_{\psi(p)}$                                                                           | Def. 11.3.2.1( $\exists I$<br>(11))    |
| 1,6             | 13 | $L_p = \mathcal{P}(D_p) \setminus \{\emptyset\}$                                                     | Th. 33.3.3.2(1, 6)                     |
| 1,2,3           | 14 | $\psi : Q_S \rightarrow Q_T$                                                                         | $\wedge E1$ (Th. 33.3.4.6(1,<br>2, 3)) |
| 1,2,3,6         | 15 | $\psi(p) \in Q_T$                                                                                    | Th. 5.2.3.9(14, 6)                     |
| 1,2,3,6         | 16 | $L'_{\psi(p)} = \mathcal{P}(D'_{\psi(p)}) \setminus \{\emptyset\}$                                   | Th. 33.3.3.2(2, 15)                    |
| 1,2,3,6         | 17 | $\mathcal{P}(D_p) \setminus \{\emptyset\} \approx \mathcal{P}(D'_{\psi(p)}) \setminus \{\emptyset\}$ | $= E(16, = E(13,$<br>12))              |
| 1,2,3,4,5,<br>6 | 18 | $\#(D_p) = \#(D'_{\psi(p)})$                                                                         | Th. 20.3.8.20(9, 10,<br>17)            |

### 3.5 Fasergrößen und profiltreue Bijektionen

**Definition 33.3.5.1** (Profilabbildung der Grundelemente).

*Halbgruppe mit Null:*  $\triangleright H_S$   
*Mengen:*  $S, s, Q_S$

$$r_S : S \rightarrow Q_S, \quad \forall s \in S \ (r_S(s) := [\{s\}]_S)$$

|     |   |                             |                   |
|-----|---|-----------------------------|-------------------|
| 1   | 1 | $H_S$                       | $A$               |
| 2   | 2 | $s \in S$                   | $A$               |
| 1   | 3 | $\text{EqRel}(K_S, \sim_S)$ | Th. 33.3.2.2(1)   |
| 2   | 4 | $\{s\} \in K_S$             | Th. 33.3.1.1(2)   |
| 1,2 | 5 | $[\{s\}]_S \in Q_S$         | Th. 11.4.2.2(3,4) |

**Theorem 33.3.5.1** (Die Profilmenge einer endlichen Halbgruppe ist endlich).

*Mengen:*  $S, Q_S, p$

$$H_S, \text{Finite}(S) \vdash \text{Finite}(Q_S)$$

|     |   |                                   |                                        |
|-----|---|-----------------------------------|----------------------------------------|
| 1   | 1 | $H_S$                             | $A$                                    |
| 2   | 2 | $\text{Finite}(S)$                | $A$                                    |
| 2   | 3 | $\text{Finite}(K_S)$              | Th. 20.3.8.18(2)                       |
| 2   | 4 | $\text{Finite}(\mathcal{P}(K_S))$ | Th. 20.3.6.17(3)                       |
| 1   | 5 | $\text{EqRel}(K_S, \sim_S)$       | Th. 33.3.2.2(1)                        |
| 6   | 6 | $p \in Q_S$                       | $A$                                    |
| 1,6 | 7 | $p \in \mathcal{P}(K_S)$          | $\wedge E1(\text{Th. 11.4.2.1}(5, 6))$ |
| 1   | 8 | $Q_S \subseteq \mathcal{P}(K_S)$  | Th. 3.5.1.1([6]:7)                     |
| 1,2 | 9 | $\text{Finite}(Q_S)$              | Th. 20.3.6.10(8,4)                     |

**Theorem 33.3.5.2 (Profilfasern als Urbilder).**

**Mengen:**  $S, p, s, C_p$

$$H_S, p \in Q_S \vdash C_p = r_S^{-1}[\{p\}]$$

|     |    |                                                                |                                               |
|-----|----|----------------------------------------------------------------|-----------------------------------------------|
| 1   | 1  | $H_S$                                                          | $A$                                           |
| 2   | 2  | $p \in Q_S$                                                    | $A$                                           |
| 1   | 3  | $r_S : S \rightarrow Q_S$                                      | Def. 33.3.5.1(1)                              |
| 2   | 4  | $\{p\} \subseteq Q_S$                                          | Th. 3.11.3.15(2)                              |
| 1   | 5  | $C_p \subseteq S$                                              | Th. 3.7.3.7(<br>Def. 33.3.2.3(1))             |
| 6   | 6  | $s \in S$                                                      | $A$                                           |
| 1,6 | 7  | $r_S(s) = [\{s\}]_S$                                           | Def. 33.3.5.1(1, 6)                           |
| 1,6 | 8  | $s \in C_p \leftrightarrow [\{s\}]_S = p$                      | Def. 33.3.2.3(1, 6)                           |
|     | 9  | $r_S(s) \in \{p\} \leftrightarrow r_S(s) = p$                  | Th. 3.11.3.9                                  |
| 1,6 | 10 | $r_S(s) \in \{p\} \leftrightarrow s \in C_p$                   | Th. 2.2.3.6(9, = $E$ (<br>Th. 2.9.1.1(7), 8)) |
| 1   | 11 | $\forall s \in S (r_S(s) \in \{p\} \leftrightarrow s \in C_p)$ | $\forall I(\rightarrow I(6, 10))$             |
| 1,2 | 12 | $r_S^{-1}[\{p\}] = C_p$                                        | Th. 5.2.5.5(3, 4, 5,<br>11)                   |
| 1,2 | 13 | $C_p = r_S^{-1}[\{p\}]$                                        | Th. 2.9.1.1(12)                               |

Für  $p \in Q_S$  bezeichne  $I_p = \{q \in Q_S \mid q \leq_S p\}$  die zugehörige untere Hauptmenge;  $I'_q$  wird entsprechend in  $Q_T$  gebildet. Die Mengen  $D_p$  sind genau ihre Urbilder unter  $r_S$ .

**Theorem 33.3.5.3 (Untere Grundelementmengen als Urbilder).**

**Mengen:**  $S, p, s, I_p, D_p$

$$H_S, p \in Q_S \vdash D_p = r_S^{-1}[I_p]$$

|     |   |                                                  |                                   |
|-----|---|--------------------------------------------------|-----------------------------------|
| 1   | 1 | $H_S$                                            | $A$                               |
| 2   | 2 | $p \in Q_S$                                      | $A$                               |
| 1   | 3 | $r_S : S \rightarrow Q_S$                        | Def. 33.3.5.1(1)                  |
|     | 4 | $I_p \subseteq Q_S$                              | Th. 3.7.3.7                       |
| 1   | 5 | $D_p \subseteq S$                                | Th. 3.7.3.7(<br>Def. 33.3.2.3(1)) |
| 6   | 6 | $s \in S$                                        | $A$                               |
| 1,6 | 7 | $r_S(s) \in Q_S$                                 | Th. 5.2.3.9(3, 6)                 |
| 1,6 | 8 | $r_S(s) \in I_p \leftrightarrow r_S(s) \leq_S p$ | Th. 3.7.2.5(7)                    |

|     |    |                                                              |                                                                                   |
|-----|----|--------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1,6 | 9  | $s \in D_p \leftrightarrow r_S(s) \leq_S p$                  | $= E(\text{Th. 2.9.1.1}(\text{Def. 33.3.5.1}(1, 6)), \text{Def. 33.3.2.3}(1, 6))$ |
| 1,6 | 10 | $r_S(s) \in I_p \leftrightarrow s \in D_p$                   | Th. 2.2.3.6(8, 9)                                                                 |
| 1   | 11 | $\forall s \in S (r_S(s) \in I_p \leftrightarrow s \in D_p)$ | $\forall I(\rightarrow I(6, 10))$                                                 |
| 1   | 12 | $r_S^{-1}[I_p] = D_p$                                        | Th. 5.2.5.5(3, 4, 5, 11)                                                          |
| 1   | 13 | $D_p = r_S^{-1}[I_p]$                                        | Th. 2.9.1.1(12)                                                                   |

**Theorem 33.3.5.4 (Korrespondierende Singleton-Profilfasern sind gleichmächtig).**

**Mengen:**  $S, T, p, q, C_p, C'_{\psi(p)}$

$$H_S, H_T, I, \text{Finite}(S), \text{Finite}(T) \vdash \\ \forall p \in Q_S \exists u (u : C_p \xrightarrow{\sim} C'_{\psi(p)})$$

*Beweis.*

Endliche geordnete Profilträger:

|          |    |                                                                            |                                               |
|----------|----|----------------------------------------------------------------------------|-----------------------------------------------|
| 1        | 1  | $H_S$                                                                      | $A$                                           |
| 2        | 2  | $H_T$                                                                      | $A$                                           |
| 3        | 3  | $I$                                                                        | $A$                                           |
| 4        | 4  | $\text{Finite}(S)$                                                         | $A$                                           |
| 5        | 5  | $\text{Finite}(T)$                                                         | $A$                                           |
| 1        | 6  | $\text{PartOrd}(Q_S, \leq_S)$                                              | Th. 33.3.2.3(1)                               |
| 2        | 7  | $\text{PartOrd}(Q_T, \leq_T)$                                              | Th. 33.3.2.3(2)                               |
| 1,4      | 8  | $\text{Finite}(Q_S)$                                                       | Th. 33.3.5.1(1, 4)                            |
| 1        | 9  | $r_S : S \rightarrow Q_S$                                                  | Def. 33.3.5.1(1)                              |
| 2        | 10 | $r_T : T \rightarrow Q_T$                                                  | Def. 33.3.5.1(2)                              |
| 1,2,3    | 11 | $\psi : Q_S \xrightarrow{\sim} Q_T$                                        | Th. 33.3.4.8(1, 2, 3)                         |
| 12       | 12 | $p, q \in Q_S$                                                             | $A$                                           |
| 1,2,3,12 | 13 | $p \leq_S q \leftrightarrow \psi(p) \leq_T \psi(q)$                        | Th. 33.3.4.7(1, 2, 3, 12)                     |
| 1,2,3    | 14 | $\forall p, q \in Q_S (p \leq_S q \leftrightarrow \psi(p) \leq_T \psi(q))$ | $\forall I(\forall I(\rightarrow I(12, 13)))$ |

Untere Urbilder und einzelne Fasern:

|                  |    |                              |                                  |
|------------------|----|------------------------------|----------------------------------|
| 15               | 15 | $p \in Q_S$                  | $A$                              |
| 1,2,3,15         | 16 | $\psi(p) \in Q_T$            | Th. 8.2.1.2(11, 15)              |
| 1,2,3,4,5,<br>15 | 17 | $\#(D_p) = \#(D'_{\psi(p)})$ | Th. 33.3.4.10(1, 2, 3, 4, 5, 15) |

|                  |    |                                                                                              |                                                           |
|------------------|----|----------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| 1,15             | 18 | $D_p = r_S^{-1}[I_p]$                                                                        | Th. 33.3.5.3(1, 15)                                       |
| 1,2,3,15         | 19 | $D'_{\psi(p)} = r_T^{-1}[I'_{\psi(p)}]$                                                      | Th. 33.3.5.3(2, 16)                                       |
| 1,2,3,4,5,<br>15 | 20 | $\#(r_S^{-1}[I_p]) = \#(r_T^{-1}[I'_{\psi(p)}])$                                             | $= E(19, = E(18,$<br>17))                                 |
| 1,2,3,4,5        | 21 | $\forall p \in Q_S \#(r_S^{-1}[I_p]) = \#(r_T^{-1}[I'_{\psi(p)}])$                           | $\forall I(\rightarrow I(15, 20))$                        |
| 1,2,3,4,5        | 22 | $\forall p \in Q_S \exists u (u : r_S^{-1}[\{p\}] \xrightarrow{\sim} r_T^{-1}[\{\psi(p)\}])$ | Th. 20.7.10(6, 7, 8, 4,<br>5, 9, 10, 11, 14, 21)          |
| 1,2,3,4,5,<br>15 | 23 | $\exists u (u : r_S^{-1}[\{p\}] \xrightarrow{\sim} r_T^{-1}[\{\psi(p)\}])$                   | $\rightarrow E(\forall E(22), 15)$                        |
| 1,15             | 24 | $C_p = r_S^{-1}[\{p\}]$                                                                      | Th. 33.3.5.2(1, 15)                                       |
| 1,2,3,15         | 25 | $C'_{\psi(p)} = r_T^{-1}[\{\psi(p)\}]$                                                       | Th. 33.3.5.2(2, 16)                                       |
| 1,2,3,4,5,<br>15 | 26 | $\exists u (u : C_p \xrightarrow{\sim} C'_{\psi(p)})$                                        | $= E(\text{Th. 2.9.1.1}$<br>(24), Th. 2.9.1.1(25),<br>23) |
| 1,2,3,4,5        | 27 | $\forall p \in Q_S \exists u (u : C_p \xrightarrow{\sim} C'_{\psi(p)})$                      | $\forall I(\rightarrow I(15, 26))$                        |

□

**Theorem 33.3.5.5 (Faserbijektionen in Urbildschreibweise).**

**Mengen:**  $S, T, p$

$$H_S, H_T, I, \text{Finite}(S), \text{Finite}(T) \vdash \\ \forall p \in Q_S \exists u (u : r_S^{-1}[\{p\}] \xrightarrow{\sim} r_T^{-1}[\{\psi(p)\}])$$

|           |    |                                                                                              |                                              |
|-----------|----|----------------------------------------------------------------------------------------------|----------------------------------------------|
| 1         | 1  | $H_S$                                                                                        | $A$                                          |
| 2         | 2  | $H_T$                                                                                        | $A$                                          |
| 3         | 3  | $I$                                                                                          | $A$                                          |
| 4         | 4  | $\text{Finite}(S) \wedge \text{Finite}(T)$                                                   | $A$                                          |
| 1,2,3,4   | 5  | $\forall p \in Q_S \exists u (u : C_p \xrightarrow{\sim} C'_{\psi(p)})$                      | Th. 33.3.5.4(1, 2, 3,<br>4)                  |
| 6         | 6  | $p \in Q_S$                                                                                  | $A$                                          |
| 1,2,3,4,6 | 7  | $\exists u (u : C_p \xrightarrow{\sim} C'_{\psi(p)})$                                        | $\rightarrow E(\forall E(5), 6)$             |
| 1,6       | 8  | $C_p = r_S^{-1}[\{p\}]$                                                                      | Th. 33.3.5.2(1, 6)                           |
| 1,2,3     | 9  | $\psi : Q_S \rightarrow Q_T$                                                                 | $\wedge E1(\text{Th. 33.3.4.6(1},$<br>2, 3)) |
| 1,2,3,6   | 10 | $\psi(p) \in Q_T$                                                                            | Th. 5.2.3.9(9, 6)                            |
| 1,2,3,6   | 11 | $C'_{\psi(p)} = r_T^{-1}[\{\psi(p)\}]$                                                       | Th. 33.3.5.2(2, 10)                          |
| 1,2,3,4,6 | 12 | $\exists u (u : r_S^{-1}[\{p\}] \xrightarrow{\sim} r_T^{-1}[\{\psi(p)\}])$                   | $= E(11, = E(8, 7))$                         |
| 1,2,3,4   | 13 | $\forall p \in Q_S \exists u (u : r_S^{-1}[\{p\}] \xrightarrow{\sim} r_T^{-1}[\{\psi(p)\}])$ | $\forall I(\rightarrow I(6, 12))$            |

Die Fasern  $C_p$  können leer sein: Ein Profil kann von einer mehrgliedrigen Teilmenge erzeugt werden, ohne das Profil irgendeiner Einermenge zu sein.

Die allgemeinen Fasersätze verlangen deshalb keine Surjektivität von  $r_S$  oder  $r_T$ . Sie verwenden das in Band 9 bereits eingeführte Auswahlprinzip zum gemeinsamen Wählen der Faserbijektionen; für den vorliegenden endlichen Indexbereich ist nur endliche Auswahl nötig.

**Theorem 33.3.5.6 (Profiltreue Bijektionen erhalten die Nullproduktrelation).**

*Mengen:*  $S, T, x, y$

$$H_S, H_T, I, f : S \xrightarrow{\sim} T, \forall s \in S (r_T(f(s)) = \psi(r_S(s))), x, y \in S \vdash xy = 0_S \leftrightarrow f(x)f(y) = 0_T$$

|             |    |                                                                              |                                                                                                     |
|-------------|----|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1           | 1  | $H_S$                                                                        | $A$                                                                                                 |
| 2           | 2  | $H_T$                                                                        | $A$                                                                                                 |
| 3           | 3  | $I$                                                                          | $A$                                                                                                 |
| 4           | 4  | $f : S \xrightarrow{\sim} T$                                                 | $A$                                                                                                 |
| 5           | 5  | $\forall s \in S (r_T(f(s)) = \psi(r_S(s)))$                                 | $A$                                                                                                 |
| 6           | 6  | $x, y \in S$                                                                 | $A$                                                                                                 |
| 4,6         | 7  | $f(x), f(y) \in T$                                                           | Th. 8.2.1.2(4, 6)                                                                                   |
| 6           | 8  | $\{x\}, \{y\} \in K_S$                                                       | Th. 33.3.1.1(6)                                                                                     |
| 4,6         | 9  | $\{f(x)\}, \{f(y)\} \in K_T$                                                 | Th. 33.3.1.1(7)                                                                                     |
| 3           | 10 | $\Phi : K_S \rightarrow K_T$                                                 | Th. 28.2.15.2( $H3$ )<br>(3)                                                                        |
| 3,6         | 11 | $\Phi(\{x\}), \Phi(\{y\}) \in K_T$                                           | Th. 5.2.3.9(10, 8)                                                                                  |
| 1,6         | 12 | $r_S(x) = [\{x\}]_S \wedge r_S(y) = [\{y\}]_S$                               | Def. 33.3.5.1(1, 6)                                                                                 |
| 2,4,6       | 13 | $r_T(f(x)) = [\{f(x)\}]_T \wedge r_T(f(y)) = [\{f(y)\}]_T$                   | Def. 33.3.5.1(2, 7)                                                                                 |
| 1,2,3,6     | 14 | $\psi([\{x\}]_S) = [\Phi(\{x\})]_T \wedge \psi([\{y\}]_S) = [\Phi(\{y\})]_T$ | Th. 33.3.4.6(1, 2, 3, 8)                                                                            |
| 5,6         | 15 | $r_T(f(x)) = \psi(r_S(x)) \wedge r_T(f(y)) = \psi(r_S(y))$                   | $\wedge I(\rightarrow E(\forall E(5), \wedge E1(6)), \rightarrow E(\forall E(5), \wedge E2(6)))$    |
| 1,2,3,4,5,6 | 16 | $[\{f(x)\}]_T = [\Phi(\{x\})]_T \wedge [\{f(y)\}]_T = [\Phi(\{y\})]_T$       | $= E(\wedge E1(12), \wedge E2(12), \wedge E1(13), \wedge E2(13), \wedge E1(14), \wedge E2(14), 15)$ |
| 2           | 17 | $\text{EqRel}(K_T, \sim_T)$                                                  | Th. 33.3.2.2(2)                                                                                     |
| 1,2,3,4,5,6 | 18 | $\{f(x)\} \sim_T \Phi(\{x\}) \wedge \{f(y)\} \sim_T \Phi(\{y\})$             | Th. 11.4.1.6(17, 9, 11, 16)                                                                         |
| 1,2,3,4,5,6 | 19 | $\{f(x)\}\{f(y)\} = z_T \leftrightarrow \Phi(\{x\})\Phi(\{y\}) = z_T$        | Th. 33.3.2.6(2, 9, 11, 18)                                                                          |
| 1,2,3,6     | 20 | $\{x\}\{y\} = z_S \leftrightarrow \Phi(\{x\})\Phi(\{y\}) = z_T$              | Th. 33.3.4.1(1, 2, 3, 8)                                                                            |

|                 |    |                                                           |                                                       |
|-----------------|----|-----------------------------------------------------------|-------------------------------------------------------|
| 1,6             | 21 | $\{x\}\{y\} = \{xy\}$                                     | Th. 33.3.1.3(<br>Ax. 33.2.1.1(1), 6)                  |
| 2,4,6           | 22 | $\{f(x)\}\{f(y)\} = \{f(x)f(y)\}$                         | Th. 33.3.1.3(<br>Ax. 33.2.1.1(2), 7)                  |
| 1,2,3,4,5,<br>6 | 23 | $\{xy\} = \{0_S\} \leftrightarrow \{f(x)f(y)\} = \{0_T\}$ | $= E(22, = E(21,$<br>Th. 2.2.3.6(20, 19)))            |
|                 | 24 | $\{xy\} = \{0_S\} \leftrightarrow xy = 0_S$               | Th. 3.11.3.18                                         |
|                 | 25 | $\{f(x)f(y)\} = \{0_T\} \leftrightarrow f(x)f(y) = 0_T$   | Th. 3.11.3.18                                         |
| 1,2,3,4,5,<br>6 | 26 | $xy = 0_S \leftrightarrow f(x)f(y) = 0_T$                 | $\leftrightarrow S(25, \leftrightarrow S(24,$<br>23)) |

In den folgenden Tabellen kürzen wir ausschließlich die ausgeschriebene Eigenschaft

$$\mathcal{K}(f) \equiv f : S \xrightarrow{\sim} T \wedge \forall s \in S (r_T(f(s)) = \psi(r_S(s)))$$

ab. Das ist die Faserverträglichkeit aus Th. 9.5.6.

**Theorem 33.3.5.7 (Rekonstruktion der endlichen Nullproduktrelation).**

**Mengen:**  $S, T, x, y, p$

$$H_S, H_T, I, \text{Finite}(S), \text{Finite}(T) \vdash \\ \exists f (\mathcal{K}(f) \wedge \forall x, y \in S (xy = 0_S \leftrightarrow f(x)f(y) = 0_T))$$

|                 |    |                                                                                              |                                                                 |
|-----------------|----|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 1               | 1  | $H_S$                                                                                        | $A$                                                             |
| 2               | 2  | $H_T$                                                                                        | $A$                                                             |
| 3               | 3  | $I$                                                                                          | $A$                                                             |
| 4               | 4  | $\text{Finite}(S)$                                                                           | $A$                                                             |
| 5               | 5  | $\text{Finite}(T)$                                                                           | $A$                                                             |
| 1               | 6  | $r_S : S \rightarrow Q_S$                                                                    | Def. 33.3.5.1(1)                                                |
| 2               | 7  | $r_T : T \rightarrow Q_T$                                                                    | Def. 33.3.5.1(2)                                                |
| 1,2,3           | 8  | $\psi : Q_S \xrightarrow{\sim} Q_T$                                                          | Th. 33.3.4.8(1, 2, 3)                                           |
| 1,2,3,4,5       | 9  | $\forall p \in Q_S \exists u (u : r_S^{-1}[\{p\}] \xrightarrow{\sim} r_T^{-1}[\{\psi(p)\}])$ | Th. 33.3.5.5(1, 2, 3,<br>4, 5)                                  |
| 1,2,3,4,5       | 10 | $\exists f \mathcal{K}(f)$                                                                   | Th. 9.5.6(6, 7, 8, 9)                                           |
| 11              | 11 | $\mathcal{K}(f)$                                                                             | $A$                                                             |
| 12              | 12 | $x, y \in S$                                                                                 | $A$                                                             |
| 1,2,3,11,<br>12 | 13 | $xy = 0_S \leftrightarrow f(x)f(y) = 0_T$                                                    | Th. 33.3.5.6(1, 2, 3,<br>$\wedge E1(11), \wedge E2(11),$<br>12) |
| 1,2,3,11        | 14 | $\forall x, y \in S (xy = 0_S \leftrightarrow f(x)f(y) = 0_T)$                               | $\forall I(\forall I(\rightarrow I(12, 13)))$                   |

$$\begin{array}{ll}
1,2,3,11 & 15 \exists f (\mathcal{K}(f) \wedge \forall x, y \in S (xy = 0_S \leftrightarrow f(x)f(y) = 0_T)) \quad \exists I(\wedge I(11, 14)) \\
1,2,3,4,5 & 16 \exists f (\mathcal{K}(f) \wedge \forall x, y \in S (xy = 0_S \leftrightarrow f(x)f(y) = 0_T)) \quad \exists E(10, [11]:15)
\end{array}$$

**Theorem 33.3.5.8 (Zwei vorgeschriebene Werte bei der Nullproduktrekonstruktion).**

**Mengen:**  $S, T, a_0, a_1, b_0, b_1, x, y, p$

$$\begin{array}{l}
H_S, H_T, I, \text{Finite}(S), \text{Finite}(T), a_0, a_1 \in S, b_0, b_1 \in T, \\
a_0 \neq a_1, b_0 \neq b_1, \Phi(\{a_0\}) \sim_T \{b_0\}, \Phi(\{a_1\}) \sim_T \{b_1\} \vdash \\
\exists f \left( \mathcal{K}(f) \wedge f(a_0) = b_0 \wedge f(a_1) = b_1 \right. \\
\left. \wedge \forall x, y \in S (xy = 0_S \leftrightarrow f(x)f(y) = 0_T) \right)
\end{array}$$

*Beweis.*

Verträglichkeit der vorgeschriebenen Werte:

$$\begin{array}{llll}
1 & 1 & H_S & A \\
2 & 2 & H_T & A \\
3 & 3 & I & A \\
4 & 4 & \text{Finite}(S) \wedge \text{Finite}(T) & A \\
5 & 5 & a_0, a_1 \in S \wedge b_0, b_1 \in T & A \\
6 & 6 & a_0 \neq a_1 \wedge b_0 \neq b_1 & A \\
7 & 7 & \Phi(\{a_0\}) \sim_T \{b_0\} \wedge \Phi(\{a_1\}) \sim_T \{b_1\} & A \\
1 & 8 & r_S : S \rightarrow Q_S & \text{Def. 33.3.5.1(1)} \\
2 & 9 & r_T : T \rightarrow Q_T & \text{Def. 33.3.5.1(2)} \\
1,2,3 & 10 & \psi : Q_S \xrightarrow{\sim} Q_T & \text{Th. 33.3.4.8(1, 2, 3)} \\
1,2,3,4 & 11 & \forall p \in Q_S \exists u (u : r_S^{-1}[\{p\}] \xrightarrow{\sim} r_T^{-1}[\{\psi(p)\}]) & \text{Th. 33.3.5.5(1, 2, 3, 4)} \\
5 & 12 & \{a_0\}, \{a_1\} \in K_S \wedge \{b_0\}, \{b_1\} \in K_T & \text{Th. 33.3.1.1(5)} \\
3 & 13 & \Phi : K_S \rightarrow K_T & \text{Th. 28.2.15.2(H3(3))} \\
3,5 & 14 & \Phi(\{a_0\}), \Phi(\{a_1\}) \in K_T & \text{Th. 5.2.3.9(13, 12)} \\
2 & 15 & \text{EqRel}(K_T, \sim_T) & \text{Th. 33.3.2.2(2)} \\
2,3,5,7 & 16 & [\Phi(\{a_0\})]_T = [\{b_0\}]_T \wedge [\Phi(\{a_1\})]_T = [\{b_1\}]_T & \text{Th. 11.4.1.5(15, 14, 12, 7)} \\
1,2,3,5 & 17 & \psi([\{a_0\}]_S) = [\Phi(\{a_0\})]_T \wedge \psi([\{a_1\}]_S) = [\Phi(\{a_1\})]_T & \text{Th. 33.3.4.6(1, 2, 3, 12)} \\
1,5 & 18 & r_S(a_0) = [\{a_0\}]_S \wedge r_S(a_1) = [\{a_1\}]_S & \text{Def. 33.3.5.1(1, 5)} \\
2,5 & 19 & r_T(b_0) = [\{b_0\}]_T \wedge r_T(b_1) = [\{b_1\}]_T & \text{Def. 33.3.5.1(2, 5)}
\end{array}$$

|           |    |                                                              |                                                                                                                                                                                                                                                                                                                                |
|-----------|----|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1,2,3,5,7 | 20 | $r_T(b_0) = \psi(r_S(a_0)) \wedge r_T(b_1) = \psi(r_S(a_1))$ | $\wedge I(= E(\text{Th. 2.9.1.1}$<br>$(\wedge E1(16)),$<br>$\text{Th. 2.9.1.1}(\wedge E1$<br>$(17)), \text{Th. 2.9.1.1}$<br>$(\wedge E1(18)), \wedge E1$<br>$(19)), = E($<br>$\text{Th. 2.9.1.1}(\wedge E2$<br>$(16)), \text{Th. 2.9.1.1}$<br>$(\wedge E2(17)),$<br>$\text{Th. 2.9.1.1}(\wedge E2$<br>$(18)), \wedge E2(19)))$ |
|-----------|----|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Faserweise Bijektion und Nullprodukte:

|                   |    |                                                                                                                                          |                                               |
|-------------------|----|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1,2,3,4,5,<br>6,7 | 21 | $\exists f (\mathcal{K}(f) \wedge f(a_0) = b_0 \wedge f(a_1) = b_1)$                                                                     | Th. 9.5.9(8, 9, 10, 11,<br>5, 6, 20)          |
|                   | 22 | $\mathcal{K}(f) \wedge f(a_0) = b_0 \wedge f(a_1) = b_1$                                                                                 | $A$                                           |
|                   | 23 | $x, y \in S$                                                                                                                             | $A$                                           |
| 1,2,3,22,<br>23   | 24 | $xy = 0_S \leftrightarrow f(x)f(y) = 0_T$                                                                                                | Th. 33.3.5.6(1, 2, 3,<br>$\wedge E1(22), 23)$ |
| 1,2,3,22          | 25 | $\forall x, y \in S (xy = 0_S \leftrightarrow f(x)f(y) = 0_T)$                                                                           | $\forall I(\forall I(\rightarrow I(23, 24)))$ |
| 1,2,3,22          | 26 | $\exists f (\mathcal{K}(f) \wedge f(a_0) = b_0 \wedge f(a_1) = b_1 \wedge \forall x, y \in S (xy = 0_S \leftrightarrow f(x)f(y) = 0_T))$ | $\exists I(\wedge I(22, 25))$                 |
| 1,2,3,4,5,<br>6,7 | 27 | $\exists f (\mathcal{K}(f) \wedge f(a_0) = b_0 \wedge f(a_1) = b_1 \wedge \forall x, y \in S (xy = 0_S \leftrightarrow f(x)f(y) = 0_T))$ | $\exists E(21, [22]:26)$                      |

□

Die beiden markierten Elemente dürfen dasselbe Profil haben. In diesem Fall werden beide Werte innerhalb derselben Faser festgelegt; die Verschiedenheit der Ausgangs- und der Zielpunkte ist genau die benötigte Voraussetzung des Zweipunktsatzes. Eine Erhaltung sämtlicher Einermengen durch  $\Phi$  ist weder vorausgesetzt noch behauptet.